

INTRODUCTION TO RESEARCH

DEFINITION OF RESEARCH

- Process of arriving at dependable solutions through planned systematic collection analysis and interpretation of data.
- A systematic process of collecting, examining, and interpreting data.

NURSING RESEARCH

A study into those aspects of professional activity which are predominantly and appropriately the responsibility of nurses. This implies:-

- Carrying out inquiry of a given phenomenon critically.
- Investigation or experimentation following some logical sequence.
- Critical analysis of existing conclusions or theories with regards to emerging facts.

TERMINOLOGIES USED IN RESEARCH

- **NURSING RESEARCH:-**Is a systematic investigation of nursing practice itself, effects of this practice on patient care or an individual family or community health situation.
- **RESEARCH DESIGN:-** Is the overall plan for collecting and analyzing data including specifications for enhancing the internal and external validity of the study.
- **CLINICAL RESEARCH:-** Is research designed to generate knowledge to guide nursing practice.
- **PILOT STUDY:-**Is a small scale version or trial run done in preparation for a major step.
- **PRE-TESTING:-** Is the trial administration of a newly developed instrument to identify imperfect areas or assess time requirements.
- **CODING:-**Is the process of transforming raw data into standardized form usually in numerical for data processing and analysis.
- **HYPOTHESIS:-** Is a statement of predicted relationship between the variables under investigations. One that states what the research is attempting to show in the study. It's a tentative answer because it is verified only after they have been tested empirically.
- **VARIABLE:-** Any factor, characteristics, quality or attribute of a person or object that varies within the population under study.
- **INDEPENDENT VARIABLE:-** It's any factor that is believed to cause or influence the dependent variable in an experimental research. It's the variable that is manipulated "cause"
- **DEPENDENT VARIABLE:-**It's the variable under observations by the investigator in order to note the effect of it on the introduction of the independent variable. It's a constant variable "effect"
- **NULL HYPOTHESIS:-** It's one that states that there is no relationship between the two variables under study.
- **EXTERNAL VALIDITY:-** It's the degree to which the results of a study can be generalized to settings or samples other than the ones under study.

- **INTERNAL VALIDITY:** - The degree to which the results can be inferred but the experimental treatment or independent variable is responsible for observed results of the dependent variable.
- **OPERATIONAL DEFINITION:** - is definition based on observable characteristic.

COMPONENTS OF RESEARCH

- Systematic-solves problems in a systematic way
- Objective-it attempts to find an unbiased solution to a problem.
- Based on observable experience or empirical evidence
- Employs carefully designed procedures or rigorous analysis.

CHARACTERISTICS OF SCIENTIFIC APPROACH IN RESEARCH

- 1) **ORDER OF SYSTEMIZATION:** - In scientific study the researcher generally moves in an orderly and systematic fashion from the definition of a problem through literature review and continuous to communicating of findings.
- 2) **CONTROL:** - Involves imposing conditions on the research situations so that biases are minimized and precision and validity are maximized.
- 3) **EMPIRICAL EVIDENCE:** - its evidence rooted in objective reality and gathered directly and indirectly through the human senses and used as the basis for generating knowledge.
- 4) **GENERALIZATION:** - The most important goal of science is to understand facts in a general broad science. The ability to go beyond the specific is an important criterion for assessing the quality of a research study.

PURPOSE OF RESEARCH

- ❖ To discover new knowledge, new facts, their correct interpretation and practical applicability {Basic research}.
- ❖ To describe a phenomenon to enable its accurate identification/understanding.
- ❖ To enable prediction of one phenomenon given another phenomenon (predictive research).
- ❖ To control or regulate phenomenon under study (experimental research), involves manipulating A to estimate its impact on B.
- ❖ To enable explanation of phenomenon, involves accurate description, predicts its occurrence & observe factors that cause its occurrence with certainty and accuracy.
- ❖ To enable theory development, formulating concepts, laws and generalization about a given phenomenon.

IMPORTANCE OF NURSING RESEARCH.

- ❖ Describe the characteristic of particular nursing situation about which little is known.
- ❖ Explain phenomena that must be considered in planning nursing care.
- ❖ Predict the problem outcome of certain nursing decision made in relation to client

care.

- ❖ Control the occurrence of undesired client outcome.
- ❖ Initiate with fair degree of confidence activities that will achieve desired behavior.

AREAS OF RESEARCH IN NURSING.

1. Nursing education.
2. Nursing administrative/management.
3. Nursing practice.

GOALS FOR CONDUCTING NURSING RESEARCH

Some of the goals for conducting nursing research include:-

- **Improvements in nursing care:-** the main reason is to foster optimum care for clients
- **Accountability for nursing practice:-** nurses must have sound rationale for their actions. This rationale should be based on knowledge that is gained through scientific research.
- **Documentation of the cost –effectiveness of nursing care**
- **Credibility of the nursing profession**

ROLES OF NURSES IN RESEARCH.

- ❖ Member of research team: - they may act as data collectors or administer the experimental intervention of study.
- ❖ User of research findings: - after evaluating research findings, nurses should use relevant findings in their practice.
- ❖ Subject in studies:-many nurses are involved in long-term survey study that is carried out.
- ❖ Principal investigator.
- ❖ Evaluator of research findings: -all nurses should be involved in the evaluation of research findings.
- ❖ Identify research problems.
- ❖ Client advocate during study: - they should be involved in making sure that the ethical aspects of search study are upheld.

REASONS FOR CARRYING OUT NURSING RESEARCH

- ❖ Study various aspects of nursing care.
- ❖ Compare 2 or more methods of care
- ❖ Evaluate use of specific care approach
- ❖ Evaluate a nursing theory

WAYS OF ACQUIRING KNOWLEDGE IN NURSING/ SOURCES OF KNOWLEDGE.

1. SCIENTIFIC RESEARCH.

Some of the facts are derived from scientific research and others are not. It is the most objective and reliable source of obtaining nursing knowledge.

2. TRADITION.

Within our culture and within nursing profession certain beliefs are accepted as truth so they are hand over from one generation to another.

3. AUTHORITIES.

These are people with specialized expertise in every field like nurses, doctors, and lawyer's e.t.c.

4. HUMAN EXPERIENCE.

We all solve problems based on prior observation and experience. Experience is a common mode of obtaining knowledge. Human beings learn through experience.

5. TRIAL AND ERROR.

Sometimes we tackle problems by successively trying out alternative solution: - method not good.

6. LOGICAL REASONING.

Deductive: - From general to specific.

Inductive: - From specific to general.

7. INTUITION.

It is the perception or explanation or insight into phenomena by instinct. It is the ability to gain new knowledge without conscious reasoning or rational process.

CAT 1:-READ AND MAKE NOTES ON HISTORY OF NURSING RESEARCH.

TYPES OF RESEARCH

a) BASED ON AIMS OF RESEARCH

1) BASIC RESEARCH

This is a type of research that aims at building knowledge but not of immediate application by practitioners.

It doesn't produce results of immediate and practical use but it provides empirical knowledge which constitutes foundations for fundamental knowledge.

2) ACTION/APPLIED RESEARCH

This is a type of research that focuses on activities in order to obtain information to solve operational problems.

It solves immediate and practical problems requiring scientific and immediate solutions on top of adding to scientific knowledge.

It provides information for planning and implementation of projects and Programmes.

It is conducted before a project is implemented (feasibility/baseline study) or to evaluate a programme (formative and summative evaluation)

Action research has 5 steps:-

1) Problem identification

Problem identification is done through various ways such as: Observation, Discussion with other people, Experience, etc.

2) Consideration of action alternatives

The researcher develops a number of alternative solutions to the problem and then considers the strengths and weaknesses of each of the solution. In evaluating the various action alternatives the research would consider such factors as; AFASS (affordability, feasibility, acceptability, sustainability, safety)

3) Action taking

This involves implementation of action that is considered the most desirable. It will include such activities as developing an action plan, mobilizing resources etc.

4) Evaluation of consequences of actions

The research after implementing the action will then review the outcomes, this will include assessment of the success of the action in addressing the problem and any challenges encountered in the implementation stage.

5) Extractions of lessons learnt

The researcher then identifies what worked well and what didn't work well, and the reasons why. This is the step that contributes to new knowledge in action research.

b) BASED ON THE TYPE OF DATA COLLECTED

1) QUANTITATIVE RESEARCH

This is a type of research that deals with quantitative data. It relies on the principle of verifiability or proof.

Knowledge is derived from what can be proven by direct observation and focuses on measurements.

Characteristics

- Concise and narrow
- Collects information under conditions of control
- Uses structured procedures and formal instruments to collect information such as questionnaires, interviews.
- Analysis is done through statistical procedures

- Investigator does not participate in the events under investigation
- Begins with preconceived ideas about how concepts are interrelated
- Emphasis objectivity in the collection and analysis of data.

Quantitative research is further divided into non-experimental and experimental research

a) EXPERIMENTAL RESEARCH

This is a study concerned with cause-effect relationships. It involves manipulation or control of independent variable (cause) and measurement of dependent variable (effect). In experimental research, the researcher is interested in controlling the extraneous variables that may influence the results.

Experimental research is further classified into:

i) True Experiments

True experiments are characterized by:-

- Manipulation of the experimental variable(s)
- Control by introduction of one or more controls over the experimental situation including the introduction of a control or comparison group. This ensures that the dependent variable is as a result of independent variable and not other variables. A control group is the group that does not receive the experimental treatment.
- Randomization – subjects are randomly assigned to either the experimental or the comparison group. Each individual has an equal chance of being placed into any of the groups in an experimental study

ii) Quasi-Experiment

This is an experimental research in which there is no control group or the subjects are not randomly assigned to groups.

It involves mostly implementing a specific treatment and later examining the effects of treatment using selected methods of measurements.

Example: randomized clinical trials.

b) NON-EXPERIMENTAL RESEARCH (DESCRIPTIVE)

This is a study that explores a phenomenon in real life situations. The purpose is to observe, describe and explore aspects of a situation as it is without manipulation. It does not try to explain or understand the underlying causes of variable of interest.

Non-experimental research is further divided into:

i) Survey

This is a type of descriptive research that focuses on obtaining information regarding prevalence, distribution, and interrelationships of variables within a population

It obtains information from a sample of people by means of self-report i.e. people in the

sample respond to a series of questions posed by the investigator.

Surveys ask subjects to report their attitudes, opinions, perceptions or behaviors.

Advantage of Survey include: ability to provide accurate information on populations using relatively small samples and also ability to obtain large amount of data rather quickly and with minimal cost.

The **disadvantage** of survey is that data maybe unreliable because in self-report, people tend to provide socially acceptable responses.

ii) Correlational studies

This is a type of research where the investigator examines the strength of relationships between variables by determining how changes in one variable are associated with changes in another variable, i.e. how variable X is related to variable Y.

This relationship can be positive or negative. The magnitude and direction of the relationship is indicated by the correlation coefficient.

In correlational studies, one group of subjects is measured on two variables.

iii) Comparative studies

This is a study that examines the differences between two intact groups on some independent variables of interest.

Unlike in experimental design the researcher is not able to manipulate the independent variable.

Comparative studies are classified as:

1) Retrospective

Dependent variable (effect) is identified in the present (a disease condition) attempt is made to determine the independent variable (cause) that occurred in the past.

Example: case-control studies.

2) Prospective

Independent variable (presumed cause is identified at the present time and then subjects are followed in future to observe the dependent variable.

Example: cohort studies.

iv) Explorative Research

It aims at exploring areas of problems, its manifestation and other factors

Differences between experimental and Descriptive Research

- Descriptive research merely describes something whereas experimental research tests a hypothesis

- Researchers can control variables in experimental research but cannot in descriptive research
- Descriptive research is subjective but experimental research is objective
- Experimental research can lead to predictions but descriptive research cannot.

3) QUALITATIVE RESEARCH

This is a type of research in which a systematic, subjective approach is used to describe life experiences and give them meaning. Data collected is in the form of words rather than numbers and data is then grouped into categories, themes and patterns.

Characteristics

- ✓ Collects information without formal structured instruments such as; KII, FGD etc.
- ✓ It involves sustained interaction with the people being studied.
- ✓ It does not attempt to control the context of the research but rather attempts to capture the context in its entirety.
- ✓ It attempts to understand the phenomenon in its entirety.
- ✓ It has few preconceived ideas and stresses the importance of people's interpretation of events and circumstances rather than the researchers interpretation
- ✓ Analyses narrative information in an organized but intuitive fashion.
- ✓ Does not aim to generalize the findings.

Advantages

- ✓ Permits research to go beyond the statistical results usually reported in quantitative research
- ✓ Enables study of attitudes and other emotional behavior

Qualitative research includes:-

i) Phenomenological studies

These are studies that examine human experiences through the descriptions that are provided by people involved. These experiences are called "lived experiences."

The goal is to describe the meaning that experiences hold for each subject.

In these studies subjects are asked to describe their experiences as they perceive them.

ii) Ethnographic studies.

This involves the collection and analysis of data about cultural groups.

Systematic process of observing, detailing, describing, documenting and analyzing the life ways or particular patterns of a culture in order to grasp the life ways/patterns of the

people in their familiar environment.

The researcher lives with the people and becomes part of their culture or interviews people who are considered knowledgeable about the culture.

iii) Grounded theory

This is a study in which data are collected and analyzed and then a theory is developed that is based (grounded) on the data. It uses purposive sampling.

iv) Historical studies

Concerned with the identification, location, evaluation and synthesis of data from the past. It seeks to discover the events of the past and to relate those events to the present and future.

v) Case studies.

This is an in-depth examination of people or groups of people, institution etc. case studies maybe considered qualitative or quantitative depending on the purpose of the study and the design chosen by the research.

In qualitative case studies, the researcher must be interested in the meaning of experiences to the subjects themselves rather than generalizing the results to other groups of people.

Case studies can be used to generate hypothesis but not to test hypothesis. They are also time consuming and costly.

c) BASED ON TIME FRAME FOR STUDY

1) LONGITUDINAL

This is a study which follows subjects over a period of time in the future. A study that follows the subjects for six or more months is considered longitudinal.

2) CROSSECTIONAL

This is a study which examines the subjects at one point in time.

THE RESEARCH PROCESS

PHASES

The research process can generally be divided into 5 phases:

a) Conceptual/thinking/planning phase

- ❖ This involves formulation and delimitation of the problem, formulation of objectives, hypothesis and research questions

- ❖ Literature review
- ❖ Development of theoretical/conceptual framework

b) The Construction phase

- Development of study methodology.
- This means the investigator clearly defines his study design, sampling, data collection, pretesting, ethical issues to be considered, data management and likely study limitations.
- Development of a work plan and a budget
- The end product of this phase is a research proposal.

c) Empirical/doing phase

This includes data collection and preparation for analysis. In this phase the research proposal is implemented.

d) Interpretive/analytic phase

This is the phase of data analysis and interpretation.

e) Communication/dissemination phase

This is the phase of writing the report and communicating the findings.

STEPS

The research process can also be divided into 10 key steps:

- 1) Identification of the research topic – This includes background information and problem statement.
- 2) Formulation of research objectives, hypothesis and research questions
- 3) Explaining the rationale/justification of the research problems.
- 4) Review of literature.
- 5) Highlighting the materials and methods.
- 6) Determining the methods of measurements.
- 7) Conducting a pretest/pilot study
- 8) Data collection and presentation
- 9) Data analysis and interpretation
- 10) Communication of the research findings.

STUDY VARIABLES

Definition

- Variables are attributes, properties and/or characteristics of individuals, events or objects that are examined in a study

OR

- Any measured characteristics on which individual observations are made

OR

- Any factor, characteristic, quality, or attribute of cases that is measured or recorded in a study

Types

1. **Quantitative variable** – This is a variable that can be measured to convey information regarding amount e.g. age, weight, height.
2. **Qualitative variable** – This is a variable that cannot be measured in amounts but in categories e.g. gender and race. It conveys information regarding attributes of a person, place or thing.
3. **Predictor/explanatory/independent**
 - o Factors that the researcher thinks explain variations in dependent variable
 - o It is the variable manipulated by the researcher to ascertain its effect if any on dependent variable
 - o Not dependent on any other
 - o Usually comes first.
4. **Dependent//criterion**
 - o A measured variable which is dependent upon the independent variable
 - o Variation on the dependent variable is what research tries to explain (effect)
 - o It is the outcome variable that the researcher wants to explain
 - o Usually there is one dependent variable.
 - o Usually comes later or last.
5. **Discrete variable**

One that has finite numbers of values between any two points e.g. 0, 1, 2, 3 or more

6. **Extraneous/intervening variable**

These are variables that are not of direct interest to the researcher but that could affect the

variables being measured. E.g. age, level of nutrition

They are thought to affect the relationship between the dependent and independent variable

7. **Continuous variable:** they have values that can be represented on a continuum without restriction to whole numbers e.g. weight between 1 and 2 pounds
8. **Dichotomous /categorical variable:-** are variables with distinct categories that do not represent a quantity e.g. gender and blood type
9. **Polychotomous Variable:** - Multiple choices Example: Preferred foods – Chinese, Japanese, American. . . .
10. **Attribute variable:** - it is a variable that are inherent characteristic of a person that the researcher measures / observes e.g. age, health beliefs or weight.
11. **Active variables:-** variables that the researcher creates e.g. salt intake variable by exposing two group of people to different amount of salt in their diets

1. IDENTIFICATION OF RESEARCH TOPIC

Research Topic/problem

- A broad issue/area that is important to health/human service profession
- An issue/area under investigation; or
- Issue or concern that puzzles the researcher due to its effects or consistence despite measures taken; e.g. neonatal deaths, maternal mortality, students' performance, etc.

Steps in topic selection

- 1) Identify what interests or puzzles one in an area, e.g. neonatal mortality
- 2) Identify keywords for the topic, i.e. zero down to the real aspect that is puzzling e. g. increase in neonatal mortality, uptake Of contraceptives
- 3) Define the topic, i.e. analyze keywords keenly by determining what to concentrate on e.g. neonatal death – early or late, malnutrition – causes or effects
- 4) Formulate the topic - title.

Characteristics of a good research topic

- Researchable/feasible – data for investigating it should be available & it should be small enough to be managed within the constraints of available resources.
- Addresses current concerns or priority areas in nursing

- Captivates the interests of the researcher
- Pay more attention to ethical issues.
- Enhance existing knowledge by filling in the gaps that exist
- Improves nursing services by contributing to more knowledge and better skills.
- Variables are clearly specified.
- Target population is clearly identified.
- Focus is clear.
- Limited to about 14 – 18 words or even fewer and redundancies avoided.

It's written on first page at the top of the page & usually not paged.

Full names of the researcher including qualifications written immediately below the title.

Award for which the proposal intended i.e. A proposal submitted in partial fulfillment for requirements of diploma (specify e.g. KRCHN), & name of awarding institution appear at the middle of the page.

Date (month & year) appear at the bottom of title page.

N/B: - All information on the title page should be in upper case and should be Centre aligned

Sources of research problems

- Experience –the nurse's everyday experience provides a rich supply of problems for investigation
- Nursing literature - ideas for research projects often come from reading the nursing literature (published research reports) and textbooks/articles
- Existing Theories – this includes theoretical schemes and conceptual frameworks that have been developed in nursing and other related disciplines.
- Ideas from external sources – a research topic can be given to students as a direct suggestion. In other cases ideas for studies may emerge as a result of brainstorming session or from discussions with other nurses, researchers, etc.
- Societal trends – social concerns reflected in the policies, legislation, and funding priorities of the Government.
- The media – issues which are frequently reported in the media can also form the basis of a research problem.
- Formal needs assessment:** - a careful assessment of needs may reveal problems that may require research to solve.

- viii. **Literature sources:** - nearly every research that has been reported concludes with some recommendations for future research.

N/B: - REPLICATION STUDY- it involves reproducing or repeating a study to determine whether similar findings will be obtained.

REPLICATION IS ESSENTIAL FOR KNOWLEDGE DEVELOPMENT FOR THE FOLLOWING REASONS.

- Establishes the credibility.
- It extends the generalizability of the findings
- Provides support for theory development.
- Decreases the acceptance of erroneous research.

CLASSIFICATION OF REPLICATION.

- i. IDENTICAL / LITERAL:** - it is an exact duplication of the original methods. Such duplication is rare, except in the case of subsequent study by the original researcher.
- ii. VIRTUAL / OPERATIONAL:** - it involves attempts to approximate the methods used in reference study as closely as possible but precise duplication is not sought.
- iii. SYSTEMATIC EXTENSION / CONSTRUCTIVE:** - methods are not duplicated but they are deliberate

Conditions/components for the existence of a research problem

- 1) There must be an individual or group which has some difficulty or problem
- 2) There must be some objectives to be attained.
- 3) There must be alternative means (or courses of action) for obtaining the objective(s) one wishes to attain
- 4) There must be some environment(s) to which the difficulty pertains.

IMPORTANT FACTORS TO CONSIDER BEFORE SELECTING A TOPIC

- ❖ Personal interest in the topic
- ❖ Importance of the research
- ❖ Time available

- ❖ Research skills
- ❖ The cost
- ❖ Ethical issues: - some topics raise ethical issues which the researcher may not easily overcome.

N/B: - Each topic should have a subject and object

STATEMENT OF THE PROBLEM

A research is usually conducted because a deficiency or a need has been identified. This deficiency or need should be clearly & specifically indicated since it justifies the research project. It can be described as the 'why' of the study.

Use reviewed literature to argue your case. Use data (tables, figures & diagrams) where possible

Writing down the research problem

A good statement:-

- Clearly and unambiguously identifies the variable(s) under consideration
- Clearly expresses the variables relationship to each other
- Specifies nature of the population(s) being studied.
- Specifies the environment(s) where the study will be undertaken.
- Restrict the scope to specific questions/ hypothesis.
- Be feasible & amenable to research: - not too wide & data for solving it available.
- Significant

A problem statement must have the following characteristics:-

- It should ask about the relationship between two or more variables
- It should be stated clearly and unambiguously, usually in question form
- It should be testable by empirical methods, that is it should be possible to collect data to answer the questions asked
- It should not represent a moral or ethical position
- The problem is stated in a question form e.g. do students continue in training programmes offering stipends?

PURPOSE

It refers to the general intention of the research. It identifies the goal of the study. It is an elaboration of the title incorporating the problem statement, target population & time of study. It should be set out in brief & specific terms. A good purpose statement should:-

- Clearly mention the central concept being explored.
- Provide general definition of the central concept.
- Give a brief summary of the method of inquiry to be used in data collection, analysis and research process.
- Specify the variables & the relationship between them.
- Describe the unit of the study.

2. FORMULATION OF RESEARCH OBJECTIVES, QUESTIONS, AND HYPOTHESIS

RESEARCH OBJECTIVES

Definition

Aspects of the phenomenon under study that the researcher desires to bring out at the end of the research period, or

Clear, concise, declarative statement expressed to direct a big study

Objectives are usually derived from the stated problem of the study.

It focuses on identification and description of variables and/or determination of the relationships among variables.

The objectives suggest the manner in which the researcher seeks to study the question and the level of evidence to be obtained through the study findings

Verbs like discover, explore or describe suggest an investigation of an infrequently researched topic. Verbs like test or compare suggest a study with better established knowledgebase.

Importance

- 1) Bridge the gap between the research and its purposes and the study design and plan for data collection and analysis
- 2) Summarize what is to be achieved by the study
- 3) Guide in formulation of testable hypothesis
- 4) Build a close link with the statement of the problem
- 5) Help the researcher to keep within the scope of study by defining the area of focus

Types

- 1) Broad
- 2) Specific

Characteristics of a good research objective

- ❖ Specific – addressing one outcome or issue

- ❖ Measurable – use action verb, e.g. determine, compare, investigate, explore, find out, examine, inquire, establish etc.
- ❖ Achievable–in terms of time, resources
- ❖ Realistic - manageable within constraints of available resources such as finances, equipment, manpower & time.
- ❖ Time bound/timely - can be accomplished within available time.

RESEARCH QUESTIONS

Definition

A Statement that identifies the key study variables, their possible interrelationships and the nature of the population of interest stated as a question.

Characteristics of a good research question

- i. It has the key variables of the study
- ii. It states the possible interrelationship of study variables
- iii. It identifies the population of study.

Components/parts

Stem–what, who, why, how

Topic

Hulley et al (2007) have suggested the use of the **FINER** criteria in the development of a good research question in general

F	Feasible	• Manageable in scope • Adequate number of participants • Adequate technical expertise • Affordable in time and money
I	Interesting	• Intrigues the investigator, peers and community
N	Novel	• New, original, unusual
E	Ethical	• Do no harm, approved by Ethics board and institution, acceptable to all
R	Relevant	• To scientific knowledge • To clinical and health policy • To future research • To current clinical practice

The **PICOT** format is a process in which clinical research questions are phrased in a manner that yields the most relevant information. The PICOT format helps;

1. In proposal development by alluding to the study population inclusion and exclusion criteria
2. To generate a question that aids in constructing the framework of the study
3. Identify an appropriate outcome measurement tool.
4. To increase the effect on the interpretation
5. To increase the applicability and generalizability of the research findings.

P	Population (patients)	What specific patient population are you interested in?
I	Intervention (for intervention studies only)	What is your investigational intervention?
C	Comparison group	What is the main alternative to compare with the intervention?
O	Outcome of interest	What do you intend to accomplish, measure, improve or affect?
T	Time	What is the appropriate follow-up time to assess outcome

HYPOTHESIS

Definition

Statement of predicted relationship between variables under study or

Tentative proposition formulated for empirical testing or

Tentative explanation for certain behavior patterns, phenomena or events that have occurred.

Sources of hypothesis

Chance

Critical analysis of the problem and discussion with other experienced nurses or colleagues

Theories - theory provides direction by stating what is known.

Observation

Intuition and Personal experience

Findings from other studies – for replication and testing

Culture

Characteristics of a workable hypothesis

1. Testable – possible to measure by either qualitative or quantitative methods.
2. Logical, factual/theory based – a sound hypothesis is consistent with existing body of theory and research findings. It should be based on a sound rationale.
3. Directly related to the research problem – hypothesis should contain essentially the same material as the problem statement or purpose statement.
4. States a relationship between independent and dependent variables.
5. Represents a single unit/subset of the problem.
6. Written in declarative sentence and in present tense.

Wording the hypothesis

A well stated hypothesis should communicate

The variables of the hypothesis

The population being studied

The predicted outcome of the hypothesis.

Purpose of hypothesis

1. Provides direction – bridges the gap between the problem and evidence needed for its solution. Identifies the anticipated direction of the relationship
2. Form the basis for the ultimate conclusions of solutions – conclusions are made on the basis of the results of the tests of hypothesis
3. Guide collection of data and provide the structure for their meaningful interpretation in relation to the problem under investigation.
4. Ensures collection of the evidence necessary to answer the question posed in the statement of the problems
5. Permits the research to understand the problem with greater clarity.

Types of hypothesis

a) Research/scientific/workable/alternative

It consists of a statement about expected relationship between variables.

It can be:

1) Directional

This is a Hypothesis that specifies the expected direction of relationship between the independent and dependent variables. Existence of the relationship is proposed and the nature of the direction of this relationship.

It uses terms such as positive, negative, less than e.g. there is a negative correlation between increase in age and performance; more girls than boys are likely to have a friend of the opposite gender before form one.

2) Non-directional

Indicate the existence of a relationship between variables but the anticipated direction of the relationship is not stated. It is formulated because:

The relationship has never been studied previously or

There has been conflicting explored findings in previous studies

b) Null/statistical

Null hypothesis states that no difference exists between groups or that there is no correlation between variables.

If in data analysis a statistically significant relationship emerges between the variables at a specified level of significance, the null hypothesis is rejected.

Rejection of the null hypothesis is the acceptance of alternative hypothesis.

Simple vs. complex

a) Simple/bivariate hypothesis

This contains only one independent and one dependent variable

b) Complex/multivariate

Complex variables contain two or more independent variables, two or more dependent variables or both.

NOTE: Unless the study is examining the relationship between variables, hypotheses are not required.

Qualitative research as well as single variable descriptive and methodological studies do not test hypotheses

When hypotheses cannot be tested in the research, research questions are posed.

ADVANTAGES OF HYPOTHESIS

- They builds researchers confidence in his results
- Sound hypothesis gives directional to the inquiry:- it helps in delimiting and singling out pertinent facts and in determining which fact may be included and which omitted
- A good hypothesis enriches theory:- after hypothesis has been verified and confirmed it becomes a part of theory

DISADVANTAGES

- It may bias the investigator towards a particular position and cause him /her lose objectivity

SIGNIFICANCE OF THE STUDY

It refers to the relevance of the study in terms of academic contributions and practical use that might be of the findings. It should consider:-

- Why and how the study will contribute to scholarly research and literature i.e. its contribution to extension of knowledge and other related studies.
- Why and how the study will help to improve practice.
- Why and how the study will improve policy.
- A study must be significant to the researched population, target population and to the nation.

THE SCOPE

It's also referred to as the delimitations of the study. It's a description of the boundary of the research in terms of content, sample size and geographical and theoretical coverage in a single line or paragraph.

3. LITERATURE REVIEW

This is a process of critically reviewing and documenting relevant published and unpublished research work done thus far in the area under investigation.

It entails the systematic identification, location and analysis of documents containing information related to the research problem being investigated.

Purpose of literature review

- i. Determine what has been done already as regards the research problem under investigation, in order to prevent duplication of work that has already been done
- ii. To refine the problem statement by establishing what others have learnt and reported
- iii. Helps the researcher to familiarize with the research methodology – by identifying the strategies procedures and measuring instruments that have been found useful in the investigation of the research problem
- iv. Helps the researcher to justify why the research is needed.
- v. To ensure that no important variable that has been found in the past to repeatedly have an impact on the problem is ignored.
- vi. Help make the researcher familiar with previous studies and thus facilitate interpretation of the results.
- vii. Help the researcher to narrow the research topic.
- viii. Examine difficulties reported by others

ix. Help determine new approaches and stimulate new ideas.

This helps to:-

- Avoid unnecessary & unintentional duplications.
- Form a framework for interpreting findings of the study.
- Demonstrate familiarity with existing body of knowledge which increases the confidence of the reader.
- To reveal strategies, procedures & measuring instruments useful in investigating the problem at hand.
- To limit the problem & define it better.
- Stimulate ideas.
- Can access recommendations for further research from past studies.
- Pulls together, integrates & summarizes what is currently known in a given area of study.

Scope of literature review

Literature review focus on areas related to the topic under study starting with the recent publications or articles (5-10years)

A minimum of 10 and maximum of 20 publications suffice for the research at this level.

When reviewing any literature the researcher needs to:

- Assess the strengths and weaknesses of past work on the subject
- Report any inconsistent findings
- Identify gaps in knowledge
- Determine the contribution of proposed study
- Consider the possibility of unintentional duplication

Types of Literature Review

1. Narrative Review (traditional review)

- o A descriptive overview of significant literature published on a given topic.
- o Critiques and summarises the literature and draws conclusions about the topic in question.
- o Brings the reader up-to-date with current literature on a topic

Narrative reviews can be;

a) Explorative

- Seeks to find out what actually exists in the academic literature in terms of theory and empirical evidence

b) Instrumental

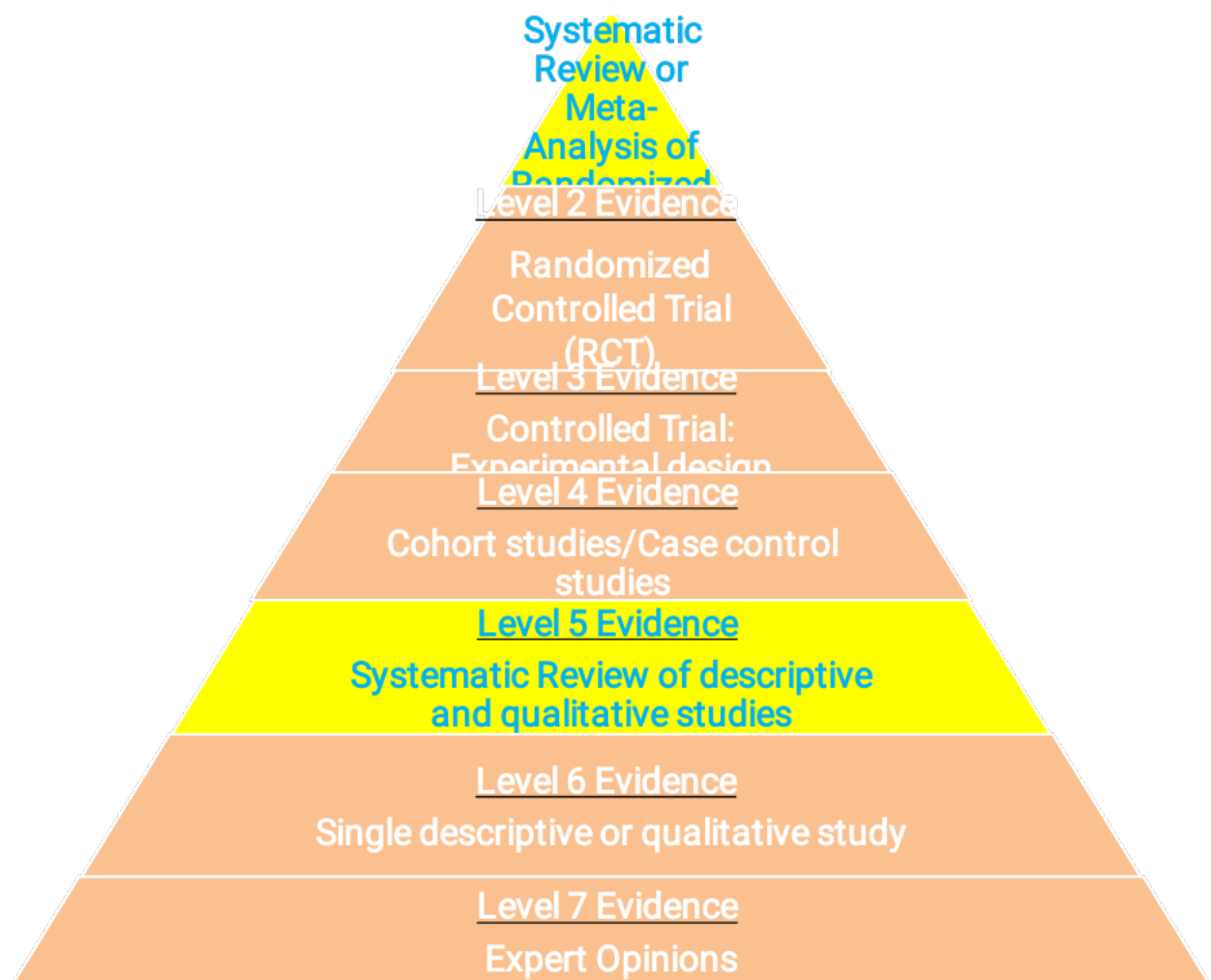
- i. Seeks to find out how to conduct research on a highly specific research problem.
- ii. It is designed to identify the best design and tools to use without having to incur unnecessary expenses.

c) Evaluative

- i. Focuses on providing a discussion of the literature in terms of its coverage, and contribution to knowledge in a particular area. It is often used to directly compare research findings of a project with other findings that are directly available

2. Systematic Review

- o A summary of evidence
- o Typically conducted by an expert or expert panel on a particular topic
- o Uses systematic and explicit methods to analyse the strength of evidence in different studies
- o Uses a rigorous process to minimize bias to identify, appraise, and synthesize studies
- o Used to answer a specific clinical question and draw conclusions about the data gathered.
- o Used to develop evidence based practices
- o Statistical methods (meta-analysis) may or may not be used to analyse and summarise the results of the included studies.
- o A Meta-Analysis is a process of using quantitative methods to summarize the results from multiple studies obtained and critically reviewed
- o The purpose of this process is to gain a summary statistic (i.e., a measure of a single effect) that represents the effect of the intervention across multiple studies.
- o A systematic review with a meta-analysis is the highest level of evidence



Sources of literature

i) Primary

This is a direct description of any occurrence by an individual who actually observed or witnessed the occurrence. It is the work written by the person who is actually involved in or is responsible for generating the idea published. This is the person who conducts empirical research and publishes it e.g. Theses and Dissertations. It's a more reliable source. Include:

-
- Diaries, journals, and letters
- Newspaper and magazine articles (factual accounts)
- Government records (census, marriage, military)
- Photographs, maps, postcards, posters
- Recorded or transcribed speeches
- Interviews with participants or witnesses (e.g., The Civil Right Movement)
- Interviews with people who lived during a particular time (e.g., genocide in Rwanda)
- Songs, Plays, novels, stories
- Paintings, drawings, and sculptures
- Published results of research studies
- Published results of scientific experiments
- Published results of clinical trials
- Proceedings of conferences and meetings

ii) Secondary sources

These include any publication written by an author who was not a direct observer or participant in the events described, e.g. books, Government documents, Journals, reports, Newspapers, Magazines.

- Biographies
- Histories
- Literary Criticism
- Book, Art, and Theater Reviews
- Newspapers
- books,
- Government documents,
- Journals,
- reports,
- Magazines
- Articles that interpret publications about the significance of research or experiments
- Analysis of a clinical trial
- Review of the results of several experiments.

Characteristics of secondary data for use

- Reliability of data
- Suitability of data
- Adequacy of data

GREY LITERATURE

- Refers to publication issued by the government, academia, business and industry, in both print and electronic format but not controlled by commercial publishing interests.

They comprises:- newsletter, reports working papers, theses, government documents, bulletins, fact sheets, conference proceedings and other publication distributed free, available subscription or for sale.

SOURCES OF INFORMATION

- ❖ Journals
- ❖ Theses
- ❖ Government documents
- ❖ Conference proceedings
- ❖ Books

- ❖ International indices
- ❖ Abstracts
- ❖ Internet

Steps

1) Identify the various published and unpublished materials available on the topic of interest and gain access to these

2) Gather the relevant information by either going through the necessary materials in a library or getting access to online sources: - **online search engines;**

1. www.cochrane.org/
2. www.rhlibrary.com
3. www.pubmedcentral.nih.gov/
4. www.nice.org.uk
5. www.google scholar.com

3) Prepare an index card or computer entry for each article read to help organize the information.

Content of an index card

- Quotation or citation
- Keyword
- A summary of content of book or article concentrating on information relevant to the study.
- Brief analysis of the content with comments such as:
- Problem that was researched
- Appropriateness of the methodology
- Important findings of the study

4) Write up the literature review

Analyze, organize and report all relevant information in an orderly way.

Use the objectives to organize your write up and use your own words using relevant references.

THEORETICAL/CONCEPTUAL FRAMEWORK

Framework

This is an explanation based on the literature you have read on how variables in the study are expected to relate to each other

Conceptual framework

Explanation based on literature and research about variables but the literature doesn't contain any particular theory that explains the relationship among variables

Theoretical framework

Variables have been studied before and have been found to be related to one another and there is an existing theory that explains the relationship or one proposed by the author

Literature should be added to explain the theory being used.

From theoretical framework:

- Testable hypothesis can be developed to examine whether the theory formulated is varied or not
- If testable hypotheses are not developed, developing a good theoretical framework is central to understanding the problem under investigation.
- Identify the network of relationships among variables considered important for the study
- Identify the problems first and then the problems that contribute to it.

Features of theoretical framework

- Contains Variables considered relevant to the study
- It shows the relationship between two or more variables and how they are related to each other
- Shows the nature and direction of the relationship as established from previous studies
- A clear explanation of why we expect the relationship to exist
- A schematic diagram of the theoretical framework so readers can see and clearly comprehend the theorized relationship.

CITATION AND REFERENCING

This is the process of identifying the sources of direct quotations and of any facts or opinions used by the researcher.

Reasons

- i. To show anyone that you understand the topic and can demonstrate your own thoughts on it.
- ii. To demonstrate that you have read widely and deeply i.e. show the depth and scope

of your research.

- iii. To enable the readers to locate where you obtained each quote or idea
- iv. By providing the original source you are acknowledging that you have read the work and recognize the original author.

Citation styles

Citation within your text link specific passages to the sources you consulted or quoted. This can be done through in text parenthetical notes, foot notes or end notes.

In addition, a list of works cited is placed at the end of the paper.

There are many different citation styles. These include:

i) Harvard University (author-date) style

This is widely accepted in academic publications. When you refer to ideas or information you have collected during your research you include the author and date of the publication referred to or cited. Also called **the author and date style**.

The name of the author and the date of publication of the work are placed in the text immediately after the work has been cited or referred to. E.g. (Gicheha 2013) or Gicheha (2013).

If two authors it will be (Gicheha and Mululu 2013)

If more than two authors add 'et al' after the first author E.g. (Gicheha et al 2013).

ii) APA style (American Psychological Association)

This is commonly used in social sciences. When using APA author date method is used in the intext citation. APA format is the official writing and publication style of the American Psychological Association (APA) and is commonly used to cite sources in psychology, education, and the social sciences. The APA citation format requires parenthetical citations within the text rather than endnotes or footnotes. Citations in the text provide brief information, usually the name of **the author and the date of publication, and sometimes page number** to lead the reader to the source of information in the reference list at the end of the paper. (Lazar, 2006, p.52). Puts the year in brackets after the author's name. Lazar, J. (2006). *Web Usability: A user-centered design approach*. Boston, MA: Pearson Addison Wesley.

iii) MLA (modern Language Association of America)

This is commonly used in literature. Does not include year or commas in intext citation (Lazar 52). Uses full first names and puts year nearer the end of entry, e.g. Lazar, Jonathan. *Web Usability. A User-Centred Design Approach*. Boston: Pearson Addison Wesley, 2006. Print.

iv) Chicago University style

footnote at the bottom of

Documentary-note styles. Put the author's name in

v) Oxford style

each page, or in an end note at the end of the assignment.

vi) Vancouver style

References are numbered sequentially in superscript (most times) as they appear in the text and are listed in that order in the reference list at the end of the text. Each reference bears the same number as the superscript number wherever it appears in the text.

Harvard University style

In intext citation more emphasis maybe placed on:

- The writer hence you start with the writer followed by the parenthesis

Kimani (2004) found out that there is a high drop out among mothers enrolled in the ANC clinic

- The idea in which case you start with the parenthesis (indirect)

There is a high drop out among mothers enrolled in the ANC clinic (Kimani, 2004)

Intext citation

- 1) Author 's name mentioned in the text

When making reference to an author's work in your text their name is followed by year of publication

E.g. Kimani (2004) found out that there is a high drop out among mothers enrolled in the ANC clinic

- 2) Author's name not mentioned in the text

The author and the year of publication should be place at the relevant point in the sentence or at the end of the sentence in brackets, e.g.

There is a high drop out among mothers enrolled in the ANC clinic (Kimani, 2004)

- 3) More than one author's work cited

Where reference is made to more than one author in a sentence and they are referred to directly, they are both cited.

E.g. Kimani (2004) and Kwamboka (2006) have both found out that there is a high drop out among mothers enrolled in the ANC clinic

OR

There is a high drop out among mothers enrolled in the ANC clinic (Kimani, 2004; Kwamboka, 2006)

4) 2, 3 or 4 authors cited

The authors are cited with the last two joined by “and”

E.g. Maina and Maranga (2004) reported that the uptake of Long-term contraceptive methods is low. Or

The uptake of Long term contraceptive methods is low (Maina and Maranga, 2004)

Maina, Makeni and Maranga (2004) reported that the uptake of Long-term contraceptive methods is low.

OR

The uptake of Long-term contraceptive methods is low (Maina, Makeni, Lijoodi and Maranga, 2004)

12. More than 4 authors cited

Only the first author should be used followed by “et al”

Maina et al (2004) reported that the uptake of Long-term contraceptive methods is low. Or

The uptake of Long-term contraceptive methods is low (Maina et al, 2004)

13. Several works by the same author in different years

If the works illustrates the same point and published in different years then the references should be cited in chronological order i.e. earliest first with years separated by semicolons

E.g. Maina (2004; 2011) reported that the uptake of Long-term contraceptive methods is low

7) Several works by one author in the same year

If several works published in the same year are referred to on a single occasion, these should be differentiated by adding a lower case letter directly with no space after the year

E.g. There is need to add life to years added to live of the elderly (Moraa, 2011a; b)

8) Corporate authors

If the work is by a recognized organization and has no personal author then it is usually cited under the body that commissioned the work. This applies to publications by Associations, Companies and Govt. Departments. It is Acceptable to use standard abbreviations or acronyms for these bodies provided that full name is given the first time with the abbreviation in brackets.

E.g. according to WHO (2006).....

9) If no author or date is given

If the author cannot be identified use anonymous (Anon) and the title of the work and date of publication. Title should be in italics

E.g. *Child Mortality in Kenya* (Anon), 2001.....

When date is not given use no date (n.d)

E.g. according to Muchai (n.d).....

10) Secondary sources

You may come across a summary of another author's work in the source you are reading which you would like to make reference to in your work.

This is cited:

Brown (1994) cited in Moraa (2006).....

11) Quoting portions of published text

If you want to include text from published work you must include quotation marks and maybe introduced by such phrases as the author states that ...

E.g. the author states that

"There is need to add life to years added to live of the elderly" (Moraa, 2011 p.20)

Start the quotation on a separate line indented 1/2" from the left margin.

12) Page numbers

Including page numbers in the reference will help the readers trace your source especially when using quotes and for paraphrasing specific paragraphs in the text

p. for single page (p.12)

pp. for multiple pages (pp.12, 40-42)

Referencing

Basic rules

- ❖ All lines after the first line of each entry in your reference list should be indented one half inch from the left margin (hanging indentation)
- ❖ Authors name are inverted (last name first). Give the last name and initials for all authors of a particular work up to the 4th author. If more than 4 authors, write the first author and the initials followed by et al.
- ❖ Reference list should be alphabetized by the last name of the first author of each work
- ❖ If you have more than one article by the same author, single author references or multiple author references with the same exact authors in exact the same order, list by the year of publication starting with the earliest.

Referencing Different sources:

a) Books

Author, initials, year. Title of the book: subtitle, Edition. Place of publication (town/city):

Publisher

E.g. Calfee, R.C (1991) APA guide to citation: a practical Guide, 2nd Edition. Washington D.C: American Psychological Association

b) Articles in Journals or periodicals

Author, initials, year. Title of the article, full title of the journal, volume number (issue number/part number) page numbers.

E.g. Imhoff, J.A (2010) Use of non-financial incentives in motivating health workers. *Journal of Human resource Management*, 66 (3), 1200-1223

Title of the journal should be in italics.

c) Electronic journal sources

(i) Journals available from Databases

Author, initials. Year. Title of the article. Full title of the journal [type of medium] volume number (issue/part number) page number. Available through: Name of Database [data accessed]

Imhoff, J.A (2010) Use of non-financial incentives in motivating health workers. *Journal of Human resource Management* [e-journal] 66 (3), 1200-1223 available through: BMC database [May 2nd, 2012]

(ii) From internet

Author, initials. Year. Title of the article. Full title of the journal [online] volume number (issue/part number) page number. Available at: full web address [data accessed]

E.g. Imhoff, J.A (2010) Use of non-financial incentives in motivating health workers. *Journal of Human resource Management* [online] 66 (3), 1200-1223 available at: <http://www.someadress.com/full/url> [May 2nd, 2012]

Instead of using URL you can use digital object identifier (DOI) when it is available

d) Articles in Magazines

Author, initials. Year. Title of the article. Full title of Magazine, Day, month, page number and column number.

Schultz, S. (2005). Cervical Cancer. The Parents, 3rd September, pp1A, 2A

f) Article in the Newspaper

Author, initials. Year. Title of the article. Full title of Newspaper, Day, month, page number and column number.

E.g. Baraza, S.A (2012) cervical Cancer. The East African Standard, 6th May, p1A, 2A

For letters to the editor, you add [letter to the editor] after the title before the title of the Newspaper.

E.g. Baraza, S.A (2012) NHIF: A Cash cow [letter to the editor] The East African Standard, 6th May, p1A, 2A

f) Interviews, email and other personal communications

No personal communication is included in your reference list. Instead, parenthetically cite in the main text only:

(Communicator's name, date of communication)

E.g. E. Robbins, Personal Communication, January 4th, 2012)

g) Conference reports.

Author/authorship, editor/organization, year. Full title of conference report, location, date, place of publication: publisher

E.g. Makali J.W, WHA, 2006. 6th Global forum on Health, Nairobi, 23rd -27th June, Geneva, WHO

h) Conference papers

Author, initials, year. Full title of conference paper in: followed by editor or name of the organization, full title of the conference, location, date, place of publication: publisher

E.g. Makali J.W, 2006. Prevalence of cervical cancer among WRA in Zambia: WHA, 6th Global forum on Health, Nairobi, 23rd -27th June, Geneva, WHO

i) DVD, video or Film

Full title of DVD or video. Year of release. [Type of medium] Director. (If relevant) Country of origin: Film studio or maker. (Other relevant details).

E. g. Great films from the 80s: a selection of clips from Warner Brothers top films from the 1980s. 2005 [DVD] New York: Warner Brothers.

For a film the suggested elements should include:

Title .Year of release. [Medium] Director. Country of origin: Film studio.

e. g. Macbeth, 1948. [Film] Directed by Orson Wells. USA: Republic Pictures.

j) Broadcast

For a broadcast the suggested elements should include:

Series title and episode name and number if relevant, Year of broadcast. [type of medium] Broadcasting organization and Channel, date and time of transmission.

e. g. Agenda Kenya, 2010. [TV programme] KBC, KBC1, 30th January 2010, 9.00 am.

For a broadcast obtained through BoB (Box of Broadcasts)

Little Britain, 2006. [TV programme recording] BBC, BBC2, 30th January 2006, 20.00 am. Available through: Box of Broadcasts database [Accessed 12th August 2011]

N/B: PLAGIARISM is the use or close imitation of the language and thought of another author and presentation of them as one's own original work. Within academia, plagiarism is considered as academia dishonest and offenders are subject to expulsion.

4. HIGHLIGHTING RESEARCH METHODOLOGY

(MATERIALS AND METHODS)

RESEARCH DESIGN

Research Design can be defined as:

The scheme, outline or plan that is used to generate answers to research problems

OR

A blue print or guide to the researcher activities

OR

An arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance with the research purpose

OR

The overall plan for obtaining answers to the research questions and for testing the research hypothesis

A research design spells out the strategies that the researcher adopts to develop information that is accurate, objective, and interpretable.

Every researcher has to identify an appropriate research design that he/she will use for the study.

THE ELEMENTS OF RESEARCH DESIGN

A good research design includes:-

- a) **Subjects:-** they are individuals who take part in the study, who will be recipients
- b) **Variables:-** they are the focus of the study and reflects the empirical aspects of the concepts being studied
- c) **Time:-** it is the frequency and the order in which observations are made.
- d) **Setting:-** it may be natural setting or laboratory depending upon study topic and researcher's choice.
- e) **Investigator's role**

Qualities of an effective Research design

- o Systematic and logical – effectively address the questions raised in the study
- o Contribute to accurate and fair interpretation of results
- o Clarify to the researcher the respondents and the means by which the study will be conducted
- o Contribute to deeper insights and better understanding of the research topic

There are 2 main research designs and within these there are several different research designs.

Before choosing a design a researcher should consider the purpose/ objectives/ hypothesis/ questions of the study.

Factors to consider when selecting a design

- Available information (state of knowledge) about the problem
- The nature of the problem and its environment (setting)

- Availability of resources for the study
- The skills and creativity of the researcher
- The type of comparisons that will be made.
- The procedures that will be used to control extraneous variables
- When and how many times data will be collected from the research subjects
- Whether an intervention will be required at the end of the study.
- Ethical consideration
- Feasibility
- Availability of subjects
- Validity of data
- Cost
- Nature of the purpose

Type of designs

1) Quantitative designs

It involves measuring data in terms of numbers.

a) Experimental design

True - determined by manipulation, randomization and control. There are three types:
-

- I. **Pretest- posttest control group design:** - it is the most used experimental design. Here (a) subjects are randomly assigned to groups, (b) a pretest is given to both groups (c) the experimental group receives the experimental treatment and the comparison group receives the routine treatment or no treatment, and the post test is given to both groups.
- II. **Posttest-only control group design:** - (a) subjects are randomly assigned to groups, (b) the experimental group receives the experimental treatment and the comparison group receives the routine treatment or no treatment and (c) a posttest is given to both groups.
- III. **Solomon four-group design:-** (a) subjects are randomly assigned to one of the four groups; (b) two of the groups, experimental group1 and comparison group 1 are pretested;(c) two of the groups, experimental group1 and experimental group 2 , receive the experimental treatment, whereas two groups, comparison group 1 and comparison group2 receive the routine treatment or no treatment; and a posttest is

given to all four groups

- **Quasi** – those in which either no comparison group or random assignment of subjects. It is a type of intervention study.
- **Pre-experimental research** - Are those in which the researcher has little control over the research. There are two types:-

One –shot case study: - a single group is exposed to an experimental treatment and observed after the treatment.

One group pretest-posttest: - it provides a comparison between a group of subjects before and after the experimental treatment.

CHARACTERISTIC OF EXPERIMENTAL RESEARCH DESIGN

- Research ideas are restricted by requirement that they be testable
- Theories and speculations are closely related to reality
- Simplicity in ideas and conceptualization is the ideal
- Research set out to test not to prove
- The concept of 'failure' is an archives interference in research activity
- The potential value of a research projects is directly related to the cogency of the questions asked
- The methods of research are intentionally devised to prevent the research are intentionally devised to prevent the researchers deluding himself and others
- Values play a legitimate and important part in research activities

b) Non-experimental design

- **Survey** – obtain data on certain variables to determine the characteristics of populations on those same variables. It collects data through self-report.

Types of surveys

- Descriptive survey
- Exploratory survey
- Explanatory survey
- Comparative survey
- Evaluative survey

ADVANTAGES

- Data are gathered from a more natural settings.
- The variable are examined as they are found in the existing social milieu
- A large amount of data can also be gathered at a fairly reasonable price
- It is the only method suitable to cover large population
- Careful pretesting of instruments and use of random sampling yields a considerable degree of representativeness.

DISADVANTAGES

- The respondents response rate may be low, thereby introducing bias
- Standardization of the questionnaire also means that the least common denominator is represented
- Unless the researcher uses the interview, which is more expensive the researcher is not able to observe the study subjects directly and loses the feel of the situation
- **Correlational studies** - examine the strength of relationships between variables.

ADVANTAGES

- An increased flexibility when investigating complex relationships among variables
- An efficient and effective method of collecting a larger amount of data about a problem area
- A potential for practical application in clinical settings
- A potential foundation for future more rigorous studies

DISADVANTAGES

- The variables of interest are beyond the researcher's control
- The researcher is unable to manipulate the variables of interest
- The researcher does not employ randomization in the sampling procedures because of dealing with pre-existing groups and therefore generalisability is decreased
- **Comparative studies** – examine the difference between intact groups on some dependent variable of interest.
- **Methodological studies** – concerned with the development, testing and evaluation of research instruments and methods.

CROSS SECTIONAL STUDY

They examine data at some point in time that is data are collected on only one occasion with the same subjects rather than on the same subjects at several points in time

ADVANTAGES

- They are less time consuming
- They are less expensive
- More manageable for the researcher because large amount of data can be collected at one point.
- The results are more readily available

DISADVANTAGES

- Subjects are unable to serve as their own control
- Economic accomplishment are sacrificed in terms of the investigators lessened ability to establish an in-depth developmental assessment of the interrelationships of the phenomena being studied.

LONGITUDINAL STUDIES

Researcher collects data from the same group at different points of time

ADVANTAGES

- Each subject is followed separately
- Increased depth of responses can be obtained and early trends in the variables of interest

DISADVANTAGES

- It is costly in terms of time ,effort and money
- Subjects may respond in a socially desirable way

CHARACTERISTIC OF QUANTINTATIVE RESEARCH

It is not flexible

1. The researcher does not necessary interact with the study participants
2. Questions are close-ended
3. They are objective. The researchers opinion does not influence the participants responses

4. They allow for study of large samples

ADVANTAGES OF QUANTITATIVE RESEARCH

- Interpretations are not influenced by the researcher
- They are useful for the study of large samples
- The numerical data obtained is easier to compare and analyze

DISADVANTAGES

- It collects narrower and sometimes superficial data.
- It is not flexible.

2) Qualitative designs

This deals with narrative form where mostly you interview people & get their view. It is concerned with in depth descriptions of people or events and data are collected through such methods as structured inter-views and participant observation. Qualitative research is not recommended for the beginning researcher. Examples include:-

- o **Case studies** – in-depth examination of people or groups of people. It may be qualitative or quantitative depending on the purpose and design chosen by the researcher.
- o **Phenomenological studies** - examine human experiences through descriptions provided by people involved (lived experiences)
- o **Ethnographical studies** – collect data from groups such as cultural groups.
- o **Grounded theory** – study that develops a theory based on the data collected.
- o **Historical studies** – concerned with collection and analysis of data from the past.

Types of evaluation in historical studies.

- a. **External criticism:** - is concerned with the genuineness of the data (validity).
- b. **Internal criticism:** - examines the accuracy of the data (reliability).

o **Multi method research**

Is the use of more than one method to get information in depth

- o **Rapid rural appraisal** – innovative research method used for collecting data in participatory research.

This approach emphasises the need for the researcher to acknowledge and appreciate that local research respondents have the knowledge and skills to be partners in the research process (Maalim, 1999). In RRA, both the researcher and the respondents collaborate and learn from each other,

getting possible solutions for the research questions (Rifkin, 1996). RRA is important for generating data from the local community such as disease patterns which they (community members) are quite familiar with. There are many RRA tools and techniques for data collection but the following are some of the more common ones

Mapping

This is drawing of sketch maps by the research respondents to identify the comparative location and importance of different resources within the research area. This may include health facilities, social amenities and infrastructure like roads, transport and means of communication (Maalim, 1999).

Daily Schedules

This involves outlining the daily activities of community members. Daily schedules are used to identify daily labour patterns and other activities (Jones, 1996:2). This tool enables the researcher to identify and discuss the daily workloads of the community members according to individuals, gender and seasons.

Venn Diagrams

This is the use of circles to indicate the relationships, importance and interactions between individuals and institutions within the research areas. Usually the respondents draw circles of varying sizes. The size of the circle and the distance from the centre indicate the importance of the institution, for example a health centre, to the respondent.

Seasonal Calendars

As the name suggests, this tool is used for gaining information from the research respondents concerning seasonal variations in patterns concerning certain issues of interest. This tool could, for example, enable the researcher to gather data on disease patterns according to the seasons.

CHARACTERISTIC OF QUALITATIVE RESEARCH

- It is typically more flexible than quantitative research
- It involves field work
- It produces rich and in depth data

- It is subjective
- The sample of study is small.
- It uses open-ended questions

Advantages

- It provides rich and in-depth data
- It is flexible
- It reduces ambiguity since the investigator can clarify issues with the subjects

Disadvantages

- The investigators opinion may influence his or her interpretation
- It is time consuming
- It is not easy to conduct with large samples.

DATA COLLECTION IN QUALITATIVE STUDIES

Semi structured interviews and participant observation are the two common types of data collection methods used. Other types include open- ended questionnaires, life histories, diaries, personal collection of letters and photographs, and official documents.

DISTINGUISHING FEATURES OF QUANTITATIVE AND QUALITATIVE RESEARCH.

QUANTITATIVE	QUALITATIVE
Focuses on relatively small number of concepts.	Attempts to understand the phenomena in its entirety.
Uses structured procedures and formal instruments to collect information.	Collects information without formal structured instruments.
Collects information under conditions of control.	Does not attempt to control the context of the research, but rather attempts to capture that context in its entirety.
Emphasizes objectivity in the collection and	Assumes that subjectivity is essential for

analysis of information.	understanding of human experience.
Analyses numeric information through statistical procedures.	Analyses narrative information in an organized, but intuitive fashion.
Incorporates logistic, deductive reasoning.	Inductive and dialectic reasoning are predominant.
Investigator does not participate in the events under investigation and is most likely to collect data from a real distance.	Involves sustained interaction with the people being studied in their own language and their own turf.

N/B

When qualitative and quantitative methods are used simultaneously, the technique is called **triangulation**

TRIANGULATION

Is a powerful technique that facilitates validation of data through cross verification from more than two sources. In particular, it refers to the application and combination of several research methodologies in the study of the same phenomenon.

- It can be employed in both quantitative (validation) and qualitative (inquiry) studies.
- It is a method-appropriate strategy of founding the credibility of qualitative analyses.
- It becomes an alternative to traditional criteria like reliability and validity.
- It is the preferred line in the social sciences.

By combining multiple observers, theories, methods, and empirical materials, researchers can hope to overcome the weakness or intrinsic biases and the problems that come from single method, single-observer and single-theory studies.

PURPOSE

The purpose of triangulation in qualitative research is to increase the credibility and validity of the results. Several scholars have aimed to define triangulation throughout the years.

TYPES

Denzin (1978) identified four basic types of triangulation

- *Data triangulation:* involves time, space, and persons
- *Investigator triangulation:* involves multiple researchers in an investigation
- *Theory triangulation:* involves using more than one theoretical scheme in the interpretation of the phenomenon
- *Methodological triangulation:* involves using more than one method to gather data, such as interviews, observations, questionnaires, and documents.

VALIDITY OF RESEARCH/ DESIGN.

Threats to internal validity.

They are factors other than the independent variable that influence the dependent variable. They include:-

- **Selection bias:** - it occurs when the subjects are not randomly assigned to experimental and comparison group.
- **History:** - it's when some events occurs during the course of study and this may influence the dependant variable.
- **Maturation:**-it may become a threat when changes occur within the subjects during an experimental study influence the study results e.g. people may become older, taller etc.
- **Testing:** - it refers to the influence of the pretest, or knowing the results of the pretest, on the post test score.
- **Instrumentation change:** - it involves the difference between the pretest and posttest measurement caused by change in the accuracy of the instruments.
- **Mortality:** - it occurs when the subject dropout rate is different between the experimental and comparison group.

Threats to external validity.

1. **Hawthorne effect:**-occurs when study participants respond in a certain manner because they are aware that they are being observed. It may also be considered a threat to internal validity.
2. **Experimenter effect:** - is a threat to the study results when the researcher characteristics or behavior influences subject behavior e.g. facial expression, age, and gender etc.
3. **Reactive effects of pretest:** - it occurs when subjects have been sensitized to the treatment through taking the pretest. So people might not respond to the treatment in the same manner if they have not received a pretest.

RESEARCH SITE

This refers to the location of the study.

The selection of a research site is essential. This is because it influences the usefulness of the information produced.

In selecting a research site:

- ✓ Identify the largest areas which are relevant to your research questions and objectives
- ✓ Consider the heterogeneity of the potential study population and choose areas or communities which represent the range of variations on the most important characteristics
- ✓ Identify and select actual communities which fulfill these criteria by making site visits, discussing with community leaders
- ✓ Issues of accessibility should also be considered.

It is important to discuss how the research site was selected and the reasons why.

E.g. the study was carried out in Mathare Slums. Purposive sampling was used to select Mathare Slums because it has.....

In selecting the study site it is important to consider the following factors:-

- Study design
- Size of the study population – availability of the adequate subjects
- Available resources
- Available time
- Purpose of the study

STUDY POPULATION

Population refers to:

Entire set of elements that meet specified criteria

OR

A complete set of persons or objects that possess some characteristics that is of interest to the researcher

OR

Entire aggregate of elements or cases that meet a designated set criteria or

A complete group/set of individuals, elements or both which have a set of common observable characteristics predefined by the investigator.

It is important for the researcher to find out as much as possible about the study population including demographics.

Types of population

1) Target population

This consists of every element in the world that meets the sampling criteria. It is the entire group of people or objects to which the researcher wishes to generalize the findings of the study.

2) Accessible population

This refers to that portion of the target population that the investigator can reasonably reach.

This is referred to as the study population and it is the population from which samples are drawn because the likelihood of getting a list of target population is low.

It is important to describe your target population in great detail discussing:

- Who they are
- Where they are located
- When they can be found
- Whether or not there is a listing of your population and whether this listing is available to you.

In describing the study population it is important to highlight the inclusion and exclusion criteria. Eligibility criteria may reflect one or more of the following issues

- Cost.
- Practical concerns.
- People ability to participate in a study.
- Design consideration.

a) Inclusion criteria

Sometimes called eligibility criteria, inclusion criteria refers to those characteristics that must be met in order to be considered for participation in the study.

The investigator specifies what the sample will look like and what characteristics all study participants will have in common.

For example: in a study of efficacy of a drug in treatment of DM, the participants must have a diagnosis of DM.

b) Exclusion criteria

This refers to those characteristics that if present, would make persons ineligible to be in the sample even though they meet all the inclusion criteria. These exclusions limit the representativeness of the sample and therefore the generalizability of the findings.

These should represent only those conditions or characteristics that might potentially make a difference in the outcome. E.g. in the above study, expectant mothers might be excluded.

SAMPLING

Is the process of selecting a portion of the population to represent the entire population.

Terminologies used in sampling:-

Sample

A chosen research group within a population

OR

A subset of cases or subjects that are taken to represent the population, **OR**

Is part of the target or accessible population that has procedurally selected to represent. It should be provided in terms of numbers & characteristics & how it was determined.

ELEMENT: - a single member of a population

REPRESENTATIVE SAMPLE: - is one whose key characteristic closely approximate those of the population

Elements of the sample

This refers to basic units about which information is collected. These are the entities that make up the samples and populations.

Sampling frame

This is a comprehensive list of all sampling elements in the target population e.g. national census register, list of all nurses

Sampling bias

Systematic over representation of some segments of the population in terms of the characteristics relevant to the research question. Occurs when samples are not carefully selected

Sampling error

This is the difference between population parameters and sample statistics. E.g. Average age. These are the errors which arise because of studying only a part of the total population.

Non sampling errors

These are errors which arise from sources other than sampling. They include errors of observation, measurement and errors of responses

Non sampling biases

They consist of biases of observation and non-observation, response biases and process biases

Total error

This is the square root of the sums of squares of variable errors and squares of biases.

Volunteers

Are subjects who approach the researcher asking to participate in the study.

THE SAMPLING PROCESS

It consists of seven sequential steps as follows:-

- Define the population
- Specify sampling frame
- Specify sampling unit
- Specify sampling method of measurements
- Determine sample size
- Specify sampling plan
- Select the sample

SAMPLING METHODS

Sampling refers to the process of selecting a number of individuals for a study in such a way that the selected individuals represent the large group (entire population) from which they are selected

Researchers use samples instead of populations because of efficiency and cost effectiveness as it is practically impossible to study the entire population

In quantitative research the major concern in sampling is how well the sample represents the target population. Therefore the resulting sample should look like the target population on important characteristics.

In qualitative studies the concern is how well the sample represents the phenomena of interest. Therefore in this case, the sample must be appropriate to provide both sufficient and relevant information on the research problem.

Sampling strategies can be classified into probability or non-probability depending on the possibility of selection of each element.

a) Probability sampling methods

This refers to random sampling strategy that gives each element in the population an equal independent chance of being selected. This is the most respected and preferred approach. It can be used to provide information about groups that are too large to study in their entity. It also provides an efficient system of capturing in a small group the variations or heterogeneity that exists in the target population.

Probability sampling techniques include:

1) Simple random sampling (SRS)

This refers to a process in which the researcher defines the population, lists and consecutively numbers all elements of the population and then randomly selects a sample from the list by:

- Putting all the numbers in a container and drawing out the desired number of elements.
- Using a table of random numbers. In this case, the researcher begins at any point on the list of numbers and reads consecutive numbers in any direction choosing those numbers that corresponds to the numbered elements in the population. This is done until the desired sample size is achieved. The starting point is selected by closing the eyes and pointing at the data in the table.

Advantages.

- Little knowledge of population is needed.
- Most unbiased of probability methods.
- Easy to analyze data and compute errors.

Disadvantages

- A complete listing of population is necessary.
- Time consuming.
- Expensive.
- Cumbersome and not the most statistically efficient method

2) Systematic/Interval sampling

Sampling technique where elements are selected from the population at predetermined fixed intervals i.e. each subject is selected at the n^{th} number of the sampling frame. The interval is determined by dividing the total population by the desired sample size.

$$\begin{array}{llll} \text{N} - \text{Total population} & & \text{i.e.} & \underline{800} \\ n \text{ No. of people wanted} & & & 40 \end{array} = 20$$

So pick a person after every 20th. Homestead.

Advantages

- Easy to draw sample.
- Economical
- Time-saving technique.

Disadvantages.

- Samples may be biased if ordering of population is not random.
- After the first sampling element is chosen, population members no longer have equal chance of being chosen.

3) Stratified sampling

This is used when the composition of the population in respect to some study characteristics important to the study is known. The population is first divided into two or more subgroups/strata by that characteristic so that each element belongs to one and only stratum. This is followed by simple random selection of elements for each group. If desired the researcher may use proportionate sampling to ensure that the sample accurately reflects the composition of the population on the characteristics by which the population is stratified.

This method is appropriate when the researcher is interested in issues related to gender, race or age disparities in the population. The goal of this technique is to achieve desired representation from various groups.

However, bias can occur in some groups being over represented.

Advantages

- Increased probability of sample being representative.

- Assures adequate number of classes for subgroups.

Disadvantage

- Requires accurate knowledge of population.
- May be costly to prepare stratified lists.
- Statistics are more complicated.

4) Cluster/multistage sampling

This strategy involves the selection of study units (cluster) instead of the study units individually. The strategy employs random selection of first larger then subsequent smaller sampling units/clusters. This method is used when:

- It is most possible to get the total list of the target population
- When the population is large and widespread.

Advantages

- Saves time and money
- Arrangements made within small number of sampling units
- Characteristic of clusters as well as those of population can be estimated

Disadvantages

- Larger sampling errors than other probability samples
- Requires assignment of each member of population uniquely to a cluster.
- Statistics are more complicated.

Advantages of probability sampling methods

- Results can be generalized
- Allows use of inferential statistics
- Sampling bias is minimized as every member has an equal chance of being selected
- Large populations can be analyzed

Disadvantages of probability sampling methods

- Non-responsive rate maybe high as the respondents are unpredictable
- Selection of the first sample member or preparation of the random table may result in a bias in the entire sample
- Requires expertise

b) Non-probability

In this method, the sample elements are chosen from the population by non-random methods. These strategies are less likely to produce sample that are representative of the population. They are used when the researcher is not interested in selecting a representative sample of the population. Mostly used in qualitative studies because the focus is on in-depth information and not generalizations or making inferences. They are widely used because of convenience and the desire to use available subjects.

They include:-

1) Convenience/accidental sampling

This is the process of selecting elements to be in the sample simply because they are readily available.

This is the simplest with potentially the least representativeness of all the sampling strategies. It is currently the most frequently used sampling strategy in nursing studies.

Advantages

- It is the cheapest and simplest
- It does not require a list of population
- It does not require any statistical expertise

Disadvantages

- It is highly biased
- It is the most reliable sampling method
- The findings cannot be generalized

2) Purposive/judgmental

Participants are handpicked by the researcher either because they are typical of the phenomenon of interest or because they are knowledgeable about the issues under investigations. Often used when the researcher desires a sample of experts.

Advantages

- It is very simple to draw
- It is less costly and involves less field work since those units can be selected which are close to each other

Disadvantages

- It is not always reliable

- It requires from the researcher considerable knowledge about the population which he usually does not possess

3) Snowball/networking

Participants already in the study are asked to provide referrals to potential study subjects. Initial study subjects are selected using purposive sampling method. Few identified then name others who have the required characteristics.

Advantages

- It is very useful in studying social groups
- It is useful for smaller populations for which no frames are readily available

Disadvantages

- It does not allow use of probability statistical methods
- It is difficult to apply this method when the population is large
- It does not ensure the inclusion of all elements in the list

4) Quota sampling

This is similar to stratified random sampling. The researcher identifies the strata of the population and then identifies the number of elements in each stratum to proportionately represent the population. Actual selection of the subjects is then done through convenience, purposive or snowballing.

Useful when the population that possess the characteristics under study is not well known and there is need to find subjects.

Advantages

- It is less costly
- It is administratively easy

Disadvantages

- The investigator selects respondents whom he knows
- It is not possible to estimate sampling errors because quota sampling does not meet the basic requirement of randomness

5) Voluntary sampling

It is a procedure in which volunteers either offer or are actively recruited to participate in a study

6) Expert sampling

It is a procedure in which the researcher selects study participants based on the need to ascertain how experts in a field would react to or judge the phenomena of interest for the study.

Advantages of non-probability sampling

- i. Useful when sampling frame/parameters of the population is not known
- ii. Doesn't require any expertise
- iii. Useful when in-depth information is required
- iv. Response rate is high
- v. Saves time and costs

Disadvantages of non-probability sampling

- i. The findings cannot be generalized
- ii. Data obtained cannot be used to test hypothesis
- iii. High sampling bias and sampling error.

CRITERIA FOR SELECTING SAMPLING TECHNIQUE

1. Purpose of the study:-if you want to generalize the findings based on the sample survey to the population
2. Measurability
3. Degree of precision
4. Information about population
5. The nature of the population
6. Geographical area of the study and the size of the population
7. Financial resources
8. Time limitation
9. Economy

SAMPLE SIZE DETERMINATION

Once the researcher has decided on who to sample and how to sample, then a decision is made on the sample size. Sample size refers to the number of subjects in a sample. Question of sample size is crucial when the objective is the ability to generalize findings to the population.

The larger the sample the smaller is the sampling error.

Factors to consider in determining the sample size

In quantitative studies, the factors that should be considered include:

1. Population factors

- Homogeneity of the population
- Expected rate of the phenomenon, event or outcome being measured. If the phenomenon occurs frequently, a smaller sample size is needed and vice-versa
- Expected attrition rate. In longitudinal studies attrition rates are expected to be high hence the need for a larger sample

2. Design factors

- Type of study
- Number of variables. As variables increase the sample size also increases
- Sampling strategy. Stratified and quota sampling requires smaller samples than simple or convenience because some of the representativeness is already built into the stratification process

3. Measurement factors

Sensitivity of the research instrument and the effect the process has on the outcome the greater the precision of the instrument, the smaller the sample since measurement error is low.

4. Practical factors

- Cost
- Convenience

Methods for determining sample size

a) Rule of 30

According to this rule there should be 30 subjects for each group or 30 subjects for each variable. This is based on the central limit theorem which states that in randomly

generated samples of 30 or more subjects, the mean characteristic will approximate the population mean

b) Power analysis

This is a statistical calculation of the number of subjects needed in order to accurately reject the null hypothesis.

Sample size can be calculated using the following **Cochran's formula**:

$$n_o = \frac{Z^2 p q}{e^2}$$

Where:

n_o is the desired sample size (for target population greater than 10,000)

z is the standard normal Deviate at the required confidence interval set at 95% (1.96)

p is the estimated proportion of the target population that has the characteristics being measured

q is the proportion of the target population estimated not to have the characteristic being measured

$$(q = 1 - p)$$

e is the level of precision set at 5% (0.05)

When there is no estimate available of the proportion of the target population who has the characteristics under study, 50% is used as p .

When the study population is less than 10,000 sample size is adjusted using the formula:

$$n = n_o / 1 + \{(n_o - 1) / N\}$$

Where:

n is the desired sample size when population is <10,000

n_o is the desired sample size when the population is >10,000

N is the estimate of the population size

N/B: we use 5- 10 % of the total population, but if the figure is too big $1/10^{TH}$ of the number is used and reasons given i.e. lack of money, time etc. The least sample size should be 50

In qualitative studies,

Relatively small non-random samples are employed because the aim of qualitative studies is to describe and analyze the meaning and experiences of particular individuals or groups.

Sample size should be determined according to the need for sufficient information to answer the research question based on the principles of data saturation and redundancy.

Participants continue to be enrolled until no new information is being uncovered (saturation). In this case redundancy occurs in all subsequent data collection encounters. This may be achieved with limited number of participants if the sample is homogenous.

ADVANTAGES OF SAMPLING

Economic in expenditure: - if data are collected for the entire population, cost will be very high

Economy in time: - the use of sampling is economical of time also.

Greater scope: - sample simplifies things and personnel with little training can collect and handle data. There is a greater scope and flexibility of studies when a sample is used

Great accuracy: - it ensures completeness and a high degree of accuracy due to limited area of operation, the volume of work is reduced, therefore careful execution of field work is possible.

Organization of convenience: - sampling involved very few organizational problems. Due to small numbers, it does not require vast facilities.

Intensive and exhaustive data: - as the number is limited it is possible to collect intensive and exhaustive data.

Suitable in limited resources: - it enables the researcher to work even with limited resources because there is a problem of resources because of poverty

Better support: - it is difficult to create rapport of the large number of people, but it is possible to develop better rapport with the respondent/subject

DISADVANTAGES

Chance of bias: - it may involve biased selection and thereby lead to draw erroneous conclusions, may be due to various reasons

Difficult in getting representative: - selection of a truly representative sample is very difficult particularly when the phenomena under study are of a complex nature

Need for specialised knowledge: - it requires the knowledge otherwise they may commit serious mistakes

Changeability of units: - since some may refuse to co-operate and some may be inaccessible and sometimes the selected cases may have to be replaced by others.

Impossibility of sampling: - sometimes the universe is too small that is it is not possible to draw a representative sample.

CHARACTERISTIC OF A GOOD SAMPLE

- It should be free from error due to bias or due to deliberate selection of the unit of the sample
- It should be free from random sampling error

- It should reproduce the characteristic of the population with the greatest possible accuracy.
- It should not suffer from incomplete coverage of the units selected from the study
- There should not be any substitution of originally selected unit by some other more convenient in any way
- Small samples properly selected may be much more reliable than large samples poorly selected

CHARACTERISTIC OF A GOOD SAMPLE DESIGN

- Sample design must result in a truly representative sample
- Sample design must be which results in a small sample error
- Sample design must be viable in the context of funds available for the research study
- Sample design must be such so that systematic bias can be controlled in better way
- Sample design should be such that the results of the sample study can be applied in general for the universe with a reasonable level of confidence

5. DATA COLLECTION METHODS AND TOOLS

Data is anything given or admitted as a fact on which a research inference will be based.

Data collection needs to be objective and systematic to ensure generalizability of the resulting outcomes to the broader population.

It is important to specify:

- What data will be collected – the type of data to be collected will also govern the other considerations. The type of data needed to answer the research questions or to test the research hypothesis should be the main consideration in data collection.
- How will data be collected – some type of research instrument is needed to gather data.
- Who will collect the data – the researcher may want to use other people outside the research team. Sometime these people are paid for their services. Training is needed to ensure data uniformity and checks made to ensure reliability of the collected data.
- Where data will be collected - optimum conditions for data collection must be sought. This includes the conditions in which the questionnaires will be filled or interviews conducted.

- When will data be collected – the determination will need to be made of the month, day, and sometimes even the hour for data collection. Also how long data collection session will take need to be determined.

Sources of data

1) Entire population

This is where the subjects earmarked for study are interviewed directly. Data from this population is referred to as **primary/new/novel data**. It is more accurate and reliable.

2) Existing records

This includes office records such as health institutions medical records, registers of births and deaths, government and non-government registers etc. Data from this source is referred to as secondary/existing/accessible data. The reliability of secondary data depends entirely on the original person who collected it.

Data collection methods

This refers to the mechanism for measuring the variable(s) of interest.

Data collection method selected depends on:

- a) The research questions or hypothesis
- b) Design of the study - The nature of the study of the subject matter
- c) Attributes of the subjects i.e. educational level
- d) Amount of knowledge available about the variable of interest - The type and depth of information to be collected
- e) Expected results and outcomes- The rate of accuracy and representative nature of the data required
- f) The unit of enquiry: - it may be an individual, household etc.
- g) The size and spread of the sample:- if it is small, interviewing may be preferable
- h) Scale of the survey:- a large scale may require mailing or interviewing through trained investigators
- i) The rate of accuracy and representative nature of the data required
- j) Time factor
- k) Availability of funds

Characteristics of data collection methods

1) Structure – research data are usually collected in a highly structured manner i.e. exactly the same information is collected from all the subjects in comparable pre-specified way

2) Quantifiability – data that will be subjected to statistical analysis must be gathered in

such a way that they can be quantified. Both quantitative and qualitative data structured and unstructured can be quantified

3) Obtrusiveness – data collection methods differ in terms of the degree to which subjects are aware of their subject status.

4) Objectivity – data collection methods strive for objectivity rather than subjectivity

Data collection tools/instruments

This refers to the devices that are used to collect information about the variable(s) under investigation. It facilitates the observation and measurement of variables of interest and depends on the data collection method(s) selected. If physiological data are sought some type of physiological instrument is needed. If observational data is sought an observational schedule or checklist is used.

Criteria for selecting a data collection instrument

1) Practicability

This concerns its cost and appropriateness for the study population in terms of the length of time required to administer the tool, physical/mental stamina required to complete the tool, special motor/physical skills required of the subjects, special training required to administer or score the instrument etc.

2) Reliability

This refers to the consistency, stability and repeatability of a data collection instrument. A reliable instrument does not respond to chance factors or environmental conditions; it will have consistent results if repeated overtime on the same person or if used by two different investigators on the same subject.

In general, the more items the instrument has the greater is the reliability.

Types of reliability

i) Stability reliability

This refers to consistency of an instrument over time

ii) Equivalence reliability

Concerns the degree to which two different forms of an instrument obtain the same results or two or more observers using a single instrument obtain the same results

iii) Internal consistency reliability/scale homogeneity

Extent to which all items on an instrument measure the same variable

3) Validity

This refers to the ability of the tool to gather data that it is intended to gather. The content of the tool is of prime importance in validity testing. The greater the validity of an instrument, the more confidence you can have that the tool will obtain data that will answer the research questions or test the research hypotheses.

Methods of establishing validity:

i) Content validity

This is concerned with the range or scope of items used to measure the variable, i.e. are the number and type of items adequate to measure the concept or construct of interest.

ii) Criterion validity

This assesses the ability of an instrument to determine subjects' responses at the present time or predict subjects' responses in the future e.g. is a suicidal potential assessment tool useful in predicting actual suicidal behavior in the future?

iii) Construct validity

This refers to the degree to which an instrument measures the construct that it is supposed to measure. Construct validity is concerned with measurement of a variable that is not directly observable but rather is a construct or abstraction derived from behavior e.g. assertiveness, anxiety

iv) Face validity: shows that it is measuring what it is supposed to measure.

v) Concurrent validity: compares an instrument's ability to obtain a measurement of subject's behavior that is comparable to some other criterion of that behavior.

A researcher can decide to use an existing instrument or develop his own.

TYPES OF ERRORS IN VALIDITY AND RELIABILITY

Random errors:- are found in validity which means that they are unsystematic in the nature and are a result of a transient state in the subject, in the context of the study or in the administration of the instrument

Systematic errors/ construct error: - occurs in reliability, it is measurement error that is attributable to relatively stable characteristics of the study, population that may bias that behavior and / or cause incorrect instrument calibration.

SOURCES OF ERROR IN DATA COLLECTION.

- Instrument inadequacies.
- Instrument administration bias.
- Environmental variation during data collection.
- Temporary subject characteristics during data collection.

Data collection methods include:-

a) Use of questionnaires

A questionnaire is a paper-and-pencil self-report instrument containing questions that respondents are asked to answer in writing.

Purpose is to collect a lot of information over a very short period of time.

Questionnaires can be used to measure knowledge levels, opinions, attitudes, beliefs, ideas, feelings, and perceptions as well as to gather factual information about respondents.

A questionnaire can be administered:-

- ✓ In person
- ✓ By mail
- ✓ Internet.

Principles in development of a questionnaire

- 1) Ensure that tool addresses the research purposes, objectives, goals and/or questions
- 2) Consider the target population to use the questionnaire so that administration is easy
- 3) Use simple familiar language without jargon and with correct grammar
- 4) Consider sequencing of questions from impersonal to personal. Less sensitive to more sensitive, broad to specific
- 5) Each statement should be short and specific and addressing a single topic only
- 6) Use neutral words to avoid leading or embarrassing the participants
- 7) Group the questions by topic and begin with questions that stimulate interest.
- 8) In wording of the questions
 - State questions in the affirmative rather than a negative manner
 - Avoid ambiguous questions - these contain words with more than one meaning or can be interpreted differently by various people
 - Avoid double negative questions - don't you disagree
 - Avoid double barreled questions – ask two questions in one

Types of questions

- ❖ Demographic questions gather data on characteristics of the subject demographic variable/attributes
- ❖ Unstructured/Open ended questions – questions that require the respondents to complete questions in their own words
- ❖ Structured/ closed ended questions - most structured type of questions. Respondents are asked to choose from given alternatives. The response categories must be exhaustive (all possible answers are given) and mutually exclusive (no overlapping).
- ❖ Contingency questions – items that are relevant for some respondents and not others. The determination of whether respondents should answer certain questions is dependent on their answers to other questions.

- ❖ Matrix questions – questions that share the same set of response categories. They are common whenever scales are used.
- ❖ Filler questions – questions that the researcher has no interest but are included in the questionnaire to reduce emphasis on specific purpose of other questions.

A cover letter should accompany a questionnaire.

Information on how to complete the questionnaire should be provided and should be clear and concise. If all questions are answered using the same format, instructions are provided at the top of the questionnaire otherwise instructions will need to precede each type of question.

Qualities of an effective questionnaire

- It is simple to understand
- Instructions are clearly given
- Questions are focused and limited to a single idea
- Each item included has a specific response and contributes to the study
- There are no leading, ambiguous or embarrassing questions
- There is a balance of questions per topic

Advantages

- Quick and generally inexpensive means of obtaining data
- One of the Easiest to test for reliability and validity
- Administration is less time consuming than interviews or observational methods
- Data can be obtained from respondents from widespread geographical areas
- Respondents can remain anonymous
- If anonymity is assured respondents are likely to give more honest answers

Disadvantages

- Mailing may be costly
- Response rate may be low
- Respondents may provide socially acceptable answers
- Respondents may fail to answer some of the items
- There is no opportunity to clarify items that may be misunderstood by respondents
- Respondents must be literate
- Respondents may not be representative of the population

FACTORS INFLUENCING RESPONSE RATE

1. Information in the cover letter
2. Neatness and clarity of the instrument.
3. Time to complete the instrument does not exceed 20-25 minutes
4. Easy of completion of the instrument.
5. Guarantee of anonymity.

Purpose of pretesting the questionnaire is to determine:

- Whether the questions as they are worded will achieve the desired results.
- Whether the questions have been placed in the best order.
- Whether the questions are understood by all classes of respondent.
- Whether additional or specifying questions are needed or whether some questions should be eliminated.
- Whether the instructions to interviews are adequate.
- Determine the human and financial resources requirement for the study.
- To determine the effectiveness of the sampling technique used.
- To determine whether the sample is representative of the population.
- To assess the time taken to administer the research instruments.
- Evaluate the procedure for data processes and analysis.
- To determine whether the proposed study is feasible.

b) Interviews

This is a data collection method that involves questioning of respondents either individually or as a group. The interviewer obtains responses from a subject in a face-to-face encounter or through a telephone call.

An interview schedule is a form with predesigned questions to be asked by the interviewer and as he gets answers, he fills them in or completes. The source of the information is referred to as the **interviewee** and the data collector is an **interviewer**. The foremost variation with the questionnaire method is the presence of the interviewer.

Questions in the interview schedule are generally open ended and responses are documented in thorough and detailed notes or transcription.

Purpose is to collect information that cannot be directly observed or is difficult to put down in writing.

Types of interview

1) Unstructured

The interviewer is given a great deal of freedom to direct the course of the interview. They are conducted like a normal conversation and topics are pursued at the discretion of the interviewer.

These are appropriate for exploratory or qualitative research studies where the researcher doesn't have enough knowledge about the topic to structure questions before data collection.

Unstructured interviews produce more in-depth information on subjects' beliefs and attitudes than any other data gathering method.

2) Structured

This involves asking a set of questions in the same order and in the same manner to all the respondents in a study

The main purpose of this method is to produce data that can be compared across respondents

3) Semi structured

An interviewer is required to ask a certain number of specific questions, but additional probes are allowed or even encouraged. Open and closed ended questions are included in a semi structured interview. In this interview data collected can be compared across all respondents. Additionally individualized data may be gathered that will provide depth and richness.

Probes are additional prompting questions that encourage the respondent to elaborate on the topic being discussed.

Respondents' answers may be entered directly on the interview schedule or recorded on audiotapes or videotapes.

Tape recording is advantageous because:

- Eye contact can be established between the interviewer and interviewee
- Total process can be captured

Disadvantages of tape recordings include

- The entire tape has to be played and transcribed during data analysis
- Respondents may be reluctant to give permission to be taped
- Written permission will be required and permission form should indicate how the information will be used and how confidentiality will be maintained.

Interviews include:

a) Key informant interview

This involves an interview between the interviewer and a person who is considered more knowledgeable in an area of study. Therefore, it relies on opinions of an individual subject or respondent.

Key informants should be interviewed individually and not as a group.

Information collected from key informants is used to validate data that has been gathered using other methods such as focused group discussions.

b) Focused group discussion

This represents coordination of an interview with 6-12 homogenous individuals who are led in the discussion of a selected topic

Each session tends to last 1-3 hours. To facilitate collection of data, each session is videotaped or audio taped for analysis at a later time. The researcher acts as a moderator. The role of the moderator includes:

- Encouraging all participants to participate in the discussion
- Stimulate discussion between participants
- Guide the group from one discussion topic to another
- Remain neutral and refrain from expressing personal opinion on a subject
- Retain control over the discussion.

The investigator of a study has the responsibility of training all the interviewers who will conduct the interviews to reduce measure variability. Training, practice and Rehearsals need to be done and the performance critiqued. The process of recording information must be explicitly communicated. Special attention should be given to the use of probes. Training should be carried out in groups so that all interviewers receive the same instructions

Role playing the interview helps the interviewer to gain some appreciation of what the actual interviews will look like. Each interviewer should play the interviewer and the respondents' part.

During interviews the researcher should seek as much privacy as possible for the interview. The interviewer and the respondents should be alone, interruptions should be avoided and television and radio should be turned off.

Interview guidelines

Before the interview:-

- The interviewer should do self-introduction, explain the purpose of the study
- Potential subjects should be told why they were chosen and how the information will be used
- The person should be told of how long the interview will take

During the interview:-

- Ensure a comfortable interview atmosphere
- Subjects should be seated in a comfortable position or lying down
- Control unnecessary noise as much as possible
- Use language easily understood by the respondents and talk in a conversation tone.
- Inform respondents that there is no right or wrong answers and no pressure should be applied to answer.
- Sensitive questions should be asked at the end of the interview-rapport is fully established

After the interview:-

- Ask if the respondents have any questions
- Further explanations of the study may be done
- Thank the respondents
- Indicate how the respondents may obtain results of the study.

Advantages

- Responses can be obtained from a wide range of subjects
- Response rate is high
- Most of the data obtained is usable
- In-depth responses can be obtained
- Nonverbal behavior and verbal mannerisms can be observed
- Data collected is quantifiable
- Allows collection of data without the requirement that the respondent can read and write

Disadvantages

- Interviews are time consuming and expensive
- Subjects may provide socially acceptable answers
- Subjects may be influenced by interviewer's characteristics

- Interviewers may misinterpret nonverbal behavior
- Researcher may miss out on some important points that are not included in the questions formulated
- Analysis of data collected is time consuming
- Subjects may be anxious because answers are being recorded.

c) Observation method

This is a method of collecting descriptive, behavioral data. It is concerned with collection of data through visual observation.

The research question or hypothesis should determine the behavior to be observed

It is important to have more than one observer to increase reliability. The training of the observers is probably the most crucial phase of the observation research to increase **interrater reliability** - the degree to which two or more observers assign the same score/rating to an observation.

Types of observation

1) Structured/controlled

A predetermined observation checklist/guide is prepared and the observer observes and records only those behaviors. This is used when the researcher has prior knowledge of the phenomenon of interest.

2) Unstructured/controlled

The observer attempts to describe the behaviors as they occur with no preconceived ideas of what will be seen. The observer observes everything and anything

Frequently, a combination of structured and unstructured observations is used.

Levels of observation

i. Complete participant

The observer takes the role of member within the sample. The data is collected via covert/hidden/undercover process. Members of the group are not informed about the data collection process

ii. Participant-as-observer

Observer continues to work from within the group but does collect the data through an overt/informed process; the members of the group are aware that the observer is taking on the dual roles of member of the group and spectator.

iii. Observer — as-participant

The observer does work from within the group but spends more time in the role of spectator instead of member of the group. Data is obtained through overt manner.

iv. Complete observer

The observer is totally in the role of complete spectator. Covert observations are used to collect the data.

- ✓ **Direct observation:-** observation of the event personally by the observer when it takes place

vi. Indirect observation:- it does not involve the physical presence of the observer, and the recording is done by mechanical, photographic or electronic devices

If covert method of data collection is to be used the researcher has to justify why because of ethical issues. On the other hand when participants know that they are being watched, the researcher has to make sure that the behavior are not modified due to this knowledge.

Advantages

- Provides highly detailed information from an external perspective of what usually occurs
 - Can be used by any individual regardless of educational preparation
 - Relatively cheap
 - Important technique for human behavior.
 - Subjects are usually available
 - Is most open to using recording devices such as video
 - Observation can begin and stop at any time
 - Instrument is quite simple to develop compared with the questionnaire

Disadvantages

- Time consuming
- Labor intensive
- Expensive
- Bias may compromise the integrity of the study
- Presence of an observer creates an artificial situation
- Extensive training is necessary if more than one observer is used to collect data.
- Events that occur rapidly or suddenly may be missed.

CHARACTERISTIC OF OBSERVATION AS A SCIENTIFIC METHOD

- There should be a limited amount of information to be collected.
- The data should be recorded systematically and analyzed over a period of time.
- The data should be congruent with the aims.
- The observation session must be planned.
- The observation and analysis must be objective.

FACTORS THAT INTERFERE WITH OBJECTIVE OBSERVATIONS

- Emotions, attitudes and values of observers may result in faulty inference
- Personal interest and commitment may color what is seen in the direction of what observers want to see.
- Anticipation of what is to be observed may affect what is observed.
- Hasty decision before adequate information is collected may result in erroneous classification

TYPES OF OBSERVATIONAL BIAS BY THE OBSERVER.

- **Enhancement of contrast effect:-** the observers distorts observations in the direction of dividing content into entities
- **Central tendency:-** occurs when extreme events are distorted toward a middle ground
- **Assimilatory:-** occurs when the observer distort observations in the direction of identity with previous input.
- **Halo effect:-** is the tendency of observers to be influenced by the characteristic in judging other unrelated characteristic.
- **Error of leniency:-** is the tendency for observers to rate everything positively.
- **Error of severity:-** is the constructing tendency to rate harshly.

d) Bio-physiological methods

This involves the collection of physical data from the subjects. These types of data are generally more objective and accurate than many of the other data collection methods. This method necessitates the use of specialized equipment to establish the physical and/ or biological condition of the subjects

Types of physiological measures

1) In vivo

This requires the use of some apparatus to evaluate one or more elements of a participant

E.g. Height, weight, BP

2) In vitro

This requires the extraction of physiological materials from the participants frequently via laboratory analysis.

This type of data collection process is frequently used with experimental and quasi experimental designs.

Advantages

- Very objective
- Precise and accurate

Disadvantages

- May be expensive
- Time consuming
- Lack of willingness of the accessible population to participate in the study
- Special expertise may be necessary to use some of the devices
- Presence of certain data collection instruments may adversely influence the subjects

e) Attitude scales

These are self-report data collection instruments that ask the respondents to report their attitudes or feelings on a continuum. They are composed of a number of related items and respondents are given a score after the item responses are totaled.

Commonly used rating scales are:

1) Likert scale (summated rating scale)

A type of composite measure of attributes that involves summation of scores on a set of items (statements) to which respondents are asked to indicate their degree of agreement or disagreement.

The scale consists of several declarative statements that express a viewpoint on a topic. The person's attitude is generally rated on a 5 point scale where:

1-strongly disagree 2- Disagree 3-undecided 4-agree 5-strongly agree

High scores indicate a favorable attitude

2) Semantic differential

This consists of a particular object, situation or event which people then rate on a series of bipolar adjectives such as good-bad, fast-slow, weak-strong.

The adjectives used are those that reflect:

- Evaluation — good-bad, helpful-unhelpful, flexible-inflexible.
- Perceived strength or potency - strong weak
- Activity - fast — slow

f) Preexisting data/records

This involves the use of information that has not been collected for research purposes. This may involve use of patient charts, records from agencies such as hospitals, letters, diaries, Family pictures etc.

The researcher has to explain what pieces of the primary data will be used in the reassessment of the information

Advantages

- Records are readily available hence data can be collected without any intrusion into peoples' lives
- Saves time
- Allows for reevaluation of alternative conclusions
- Researcher can identify trends because the entirety of the incident can be manipulated
- Less expensive

Disadvantages

- Data source is restricted since only the data that was initially amassed can be used
- Potential for data to be outdated
- Restriction on accessing certain documents such as those controlled to protect individual privacy
- Original sampling, inclusion and exclusion criteria could negatively affect secondary assessment of archived data
- Lack of open ended or qualitative data related to the question.

g) DOCUMENT ANALYSIS (CRITICAL INCIDENT TECHNIQUE)

This is the critical examination of public or private recorded information related to the issue under investigation.

It enables the researcher to access data at his convenient time hence saving time & money in transcribing.

h) THE DELPHI TECHNIQUE

It's a device for tapping the ideas of a knowledgeable group of people by mail survey. The purpose is usually to arrive at a consensus of opinion on an important subject for the purpose of predicting the future, developing program planning or for project evaluation.

The procedure includes the following steps:-

- i) Selecting a representative panel that agrees to participate in the survey.
These are contacted personally or by mail to explain about the survey & request their participation.
- ii) Get opinion on the panel members of the subject under study. They are requested to list a specified number of ideas or opinions. These opinions are then combined & returned to the panel members who are asked to select a specified number of the most important ones. Later the opinions may be further reduced to limited number of the most important.

This technique provides a way of obtaining opinions of a large group of experts & obtaining consensus upon them without face to face contact.

It preserves anonymity, provides controlled feedback & control group consensus.

i) Q METHODOLOGY

It is also called **Q-SORT** is a means of obtaining data in which subjects sort statements into categories according to their attitudes towards, or rating, of the statements. Subjects are presented with a number of words or statements that are written on cards or pieces

j) PSYCHOLOGICAL TEST

These are method used to assess the personality characteristic of people. The most commonly methods are:-

a). Personality inventories: - are self – report measures used to assess the differences in personality traits, needs, or values. They seek information about a person by asking questions or requesting response to statements that are presented. They are grouped as norm or average and then scores are awarded.

b). Projective technique: - a subject is presented with stimuli that are designed to be ambiguous or to have no definite meaning. Then the person is asked to describe the stimuli

or tell what they appear to represent. The responses reflect the internal feelings of the subjects that are projected onto external stimuli. It is the best method of collecting physiological data. They prevent subjects from providing socially acceptable responses to questions.

THE APPROACH OF COLLECTING DATA

1. When approaching a new interviewee or household do self-introduction & show your identification.
2. Explain carefully why you have come & the purpose of the survey.
3. If explanations are understood seek permission to start interviewing.
4. Establish rapport during the interview (by good listening, observing non-verbal communication & being sensitive to others feelings)
5. Fill all the responses at the time they are given.
6. Keep forms or carry them in a file or folder. Use pencil to fill the forms.
7. Any measuring instrument should be carried in a pack or case for convenience.
8. Make a daily check on the condition of the equipment to ensure it's kept clean & in good working order.
9. Before you leave the individual recheck all the forms to verify that all the responses have been filled in.

All lab work specimens should be numbered properly.

10. At the end of each day all forms that have been filled should be checked by somebody else. Any data missing can be filled the next day by returning to the appropriate person & complete or collect the information.
11. After checking the forms tallying the results can be done each day.

MEASUREMENT

Is the central to the process of obtaining data and it should consider the following aspects:
- objectivity, consistency and validity.

Measurement: - is the process of assigning numbers to variables

LEVEL OF MEASUREMENT

There are four levels of measurement i.e.

- I. **Nominal level of measurement:** - objects or events are named or categorized. The categories must be distinct from each other. Examples are gender, religion, marital status and political party. It is usually considered as the lowest level of the measurement levels.
- II. **Ordinal level of measurement:** - data is ranked and categorized. For example anxiety levels of groups of people might be categorized as mild, moderate, and severe.
- III. **Interval level of measurement:** - the distance between the ranks can be specified e.g. numbers on the scale of thermometer.
- IV. **Ratio level of measurement:** data can be categorized and ranked, the distance between the ranks can be specified and in addition a true or natural zero point can be identified.

Sources of error in measurement

Respondent: - at times the respondent may be reluctant to express strong negative feelings

Situation: - situational factors may also come in the way of correct measurement

Measurer: - the interviewer can distort responses by rewording or reordering question

Instrument: - error may arise because of the defective measuring instrument.

CONDUCTING A PRETEST/PILOT STUDY

Pilot study- a small version of a proposed study conducted to refine the methodology. It involves going through the entire research procedure with a small sample. It uses the same respondents and setting as the main study.

Purpose

- i. To determine whether the proposed study is feasible
- ii. Identify any problems with the research design
- iii. To ensure that items in the data collection instrument are stated clearly and have the same meaning to all research respondents.
- iv. To assess the time taken to administer the research instrument
- v. To determine whether the sample is a representative of the population
- vi. To determine the effectiveness of the sampling technique used
- vii. To give researcher the real experience in the field

- viii. To determine the human and financial resources required for the study.
- ix. To determine the effectiveness of the training to research assistants where necessary
- x. To evaluate the procedure for data processing and analysis.

Ideally a pretest should be done before the full scale study is done i.e. before finalizing the proposal.

Procedure for conducting a pretest:

- i.* Ask colleagues to review the questions critically- to identify if the questions are clear and whether they meet study objectives.
- ii.* Pretest the questionnaire on people who are very similar to your target group
- iii.* Simulate the actual data collection procedure. If you will use assistants, they should also be involved in pretesting
- iv.* Obtain feedback about the form and content of the questionnaire
- v.* Check if questions produce the information needed.
- vi.* Try out tabulations and analysis procedure to determine if the questionnaire yields data that can be analyzed
- vii.* Revise the tools.

DATA COLLECTION PROCESS

Data collection refers to the process of gathering information to serve or prove some facts. This will determine the findings of the entire research study.

Purpose of data collection

- To stimulate new ideas because data collection helps in identifying areas related to the research topic that need improvement or further evaluation
- To highlight a situation and there create awareness and improvement
- To influence legislative policies or regulations
- To provide justification for an existing programme or need for a new programme

- It is the only reliable way to evaluate the responsiveness and effectiveness of the study

It promotes decision making and resource allocation that are based on solid evidence

Data collection plan

Before beginning data collection process, a plan should be drawn up. This will ensure that:

- One has a clear overview of what tasks have to be carried out, who should perform them and the duration of the tasks
- Human and material resources for data collection can be organized in the most efficient way
- Problems that will require modifications to the proposal are identified.

Steps in developing a plan:

- 1) List the tasks that have to be carried out and who will be involved
- 2) Make a rough estimate of the time needed for different parts of the study.
- 3) Identify the most appropriate period in which to carry out the research
- 4) Schedule the different activities that have to be carried out each week in a plan.

In developing a plan the **following factors should be considered:**

- Type of data to be collected and demands of the project
- Availability of research team members and research assistants
- Appropriate season(s) to conduct the field work
- Accessibility and availability of the sample population
- Public holidays and vacation periods.

Stages of data collection

1. Permission to proceed

Consent must be gotten from relevant authorities, individuals and the community in which the study is to be carried out. This may involve organizing meetings at National/County/Sub-county/Institutional and at the village level.

2 Data collection

When collecting data it is important to consider the logistics. Identify who will collect what, when and with what resources.

When allocating tasks for data collection, it is important that tasks should be listed, then identify who can best implement each task.

Use of research assistants:

Where research assistants have to be used:

- They should be from the same educational level
- They should be knowledgeable concerning the topic and local conditions but not object of the study themselves
- They should be adequately trained
- Ongoing supervision should be arranged especially if the area to be covered is large.

It is important to devise methods to ensure **data quality** by:

- Requiring interviewers to check whether the questionnaire is filled before finishing each interview.
- Asking supervisors to check at the end of each day whether the questionnaires are filled completely and whether recorded information makes sense
- Having the researchers review the data during the data analysis stage to check whether data are complete and consistent.

3 Data handling

Once data has been collected, a clear procedure should be developed for handling and storing them. During this stage:

- a) Check that all the data are uniform, complete and accurate
- b) At some stage number the questionnaires either at the time of data collection or when questionnaires are stored
- c) Identify the person responsible for storing data and the place where it will be stored
- d) Decide how data should be stored either:

- Paper storage - where coded data is written on paper before analysis
- Electronic storage - using a computer.

Sources of error in Data Collection

- i) Instrument inadequacies. This is concerned with the items used to collect data and the instructions to the subjects that are contained in the instrument.
- ii) Instrument administration biases. This may occur if the instrument is not administered in the same fashion to all subjects.
- iii) Environmental variations during data collection
- iv) Temporary characteristics of the subjects during data collection such as levels of anxiety, tiredness, influencing responses etc.

ETHICAL CONSIDERATION

- Ethics is a branch of philosophy which deals with one's conduct and serves as a guide to behavior.
- The issue of ethics is very important to research & despite high value of knowledge gained through research, knowledge cannot be pursued at the expense of human dignity.
- A researcher should describe how he or she will ensure that ethical requirements are upheld in the study.
- Major ethical issues of concern are informed consent, privacy and confidentiality, anonymity and researchers responsibility.

ETHICAL ISSUES IN NURSING RESEARCH.

- Researcher must inform the subject s about the research.
- Research must be for the good of the society.
- Research must be conducted on animal experiment if possible.
- Researcher should avoid injury to research subject.
- Researcher must be qualified to conduct the research.
- Both the researcher and subject should be able to stop the research study if problems occur.

ETHICAL PRINCIPLES IN DATA COLLECTION

There are three principles that guide researchers each with rights that need to be observed

1) Principle of beneficence

This is a fundamental principle that seeks to protect subjects from harm, exploitation and maximize benefits. This principle includes:

- i) **Freedom from harm**-exposing subjects to experiences that result in serious or permanent harm is unacceptable.
 - Only scientifically qualified persons should conduct research especially if potentially dangerous equipment or specialized procedure are used.
 - Avoid inflicting psychological harm by phrasing questions carefully, debriefing and providing the researcher's contact.
- ii) **Freedom from exploitation**- Subjects should not be placed at disadvantages or exposed to situations for which they have not been explicitly prepared.
 - Therefore, subjects should be explained that their participation or information they might provide would not be used against them in any way.
 - The subjects enter into a special relationship with the researcher. This relationship should not be overtly or maliciously exploited.
- iii) **Risk/benefit ratio**- Researchers are expected to carefully assess the risks and benefits that would be incurred in conduct of a study before its inception both to the researcher and the subjects.

Research should not be undertaken when perceived risks outweighs the anticipated benefits of the research.

2) Principle of respect for human dignity

i) Right to self determination

- Prospective subjects have the right to decide voluntarily whether or not to participate in a study without the risk of incurring any penalties or prejudiced treatment.
- Subjects have the right to decide at any point to terminate their participation , to refuse to give information or to ask for clarifications or specific questions
- Humans should be treated as autonomous agents capable of controlling their own activities and destinies

- Freedom from coercion of any type.

ii) Right to full disclosure- The researcher should fully describe the nature of the study, the subject's right to refuse participation, the researcher's responsibility and likely risks and benefits so that the subjects can make informed voluntary decision about their participation in a study.

iii) Informed consent

This implies that these search subjects /interviewees /participants /observant / respondent have been informed about the procedures of a study in which they have been asked to participate.

The researcher should provide information on purpose of research, expected duration, any unforeseen risks/ discomforts to the participants, benefits, extent of privacy & confidentiality.

Elements of informed consent:

- Participant's status .e.g. " I am a nursing student at-----conducting a research study as part of the requirements for-----"
- The subjects have to be adequately informed regarding the research i.e. the type of information to be obtained from them
- Degree of understanding of the subjects to be capable of comprehending the information
- The fact that the subject have the power of free choice enabling them to consent voluntarily to participate in the research or not.

Preconditions for informed consent:

- ✓ Competence to understand and decide
- ✓ Voluntary willingness in deciding without coercion.

Informed consent is based on the right to full disclosure and right to self-determination. Consent form documents the consent process

Vulnerable subjects

These are special subjects who need additional procedures since they cannot give fully informed consent or may be at a higher risk of unintended side effects due to their circumstances. These include:

- Mentally or emotionally disabled people
- Physically disabled people

- Children
- Pregnant women

3) Principle of justice

- i) **Right to fair treatment**-Subjects have the right to fair and equitable treatment before, during and after their participation in the study.
- ii) **Right to privacy**- Subjects' privacy should be maintained throughout the study. To ensure this:
 - Strict confidentiality of all data collected should be ensured
 - Anonymity should also be maintained

- A promise of confidentiality is guaranteed to subjects that information will not be made public or accessible to parties other than those involved in the study.

Importance of ethics in research:

- To ensure that human subjects are not harmed
- To ensure that human subjects are not wronged by being treated as mere means or objects
- Access to research may provide benefits to participants that are otherwise unavailable
- Health benefits to society results from research.

DATA ANALYSIS AND PRESENTATION

Data analysis

This is the process of examining what has been collected in a survey or experiment and making deductions and inferences.

This deals with the organization, interpretation & presentation of collected data.

Data processing

This refers to the transfer of quantified data on computer storage to facilitate electronic computation of data.

Once data has been entered, the choice of statistical procedure will depend on:

- Proposed hypothesis and objectives- for exploratory study, descriptive analysis is adequate. If hypothesis will be tested, inferential statistics will be used.
- Research design- experimental design which comprises of various groups calls for analysis of variance
- Type of measurement scale used

We should think about data analysis upfront rather than waiting until all the data has been collected. This will enable us collect the right data for analysis.

Types of analysis

***a)* Quantitative analysis**

Steps

***i)* Data organization**

This is done immediately the tools have been received from the field.

All the tools are checked for accuracy, completeness and uniformity. Those that are incomplete or do not make sense are left out. A record of the decision should be kept.

A unique identifier such as a serial number is assigned to the remaining questionnaires. The data is then entered manually or by computer.

***ii)* Coding**

This is the process of assigning numerals or other symbols to answers so that responses can be put into a limited number of categories or classes. The classes should be appropriate to the research problem under consideration. The coding process starts with the preparation of a code book.

***iii)* Classification/categorization**

This is grouping of related facts into classes according to some predetermined characteristics.

Data classification is important because:

- It facilitates comparison
- It gives prominence - top the important information gathered while dropping out unnecessary elements
- It enables statistical treatment of materials collected
- It pinpoints the most significant features of data at a glance

***iv)* Tabulation**

This is the process of summarizing raw data and displaying the same in compact form. Tabulation is essential because:

- It conserves space and reduces explanatory and descriptive statements to minimum
- It facilitates process of comparison
- It facilitates the summation of items and the detection of errors and omissions
- It provides a basis for various statistical comparisons. It involves calculation of mean, median mode etc.

The type of calculations to use depends upon what you want to know

Do you want the proportion of people who answered in a certain way?	Percentage
Do you want the average number or average score?	Mean
Do you want the middle value in a range of values or scores?	Median
Do you want to show the range in answers or scores?	Range
Do you want to compare one group to another?	Cross tab
Do you want to report changes from pre to post?	Change score
Do you want to show the degree to which a response varies from the mean?	Standard deviation

v) Interpretation

This is the process of attaching meaning to data. It is here that inferential deductions are made from the data from which it would help in testing the hypothesis and thus solving the research problem

It is done within the frame of reference of the research problem and hypothesis.

Interpretation requires careful judgments because often the same data can be interpreted in different ways.

Some of the techniques for interpreting data include:

- Extending the analysis by raising questions
- Connect findings to personal experiences
- Seek the advice of critical friends, colleagues
- Contextualize findings in the research
- Turn to theory

Common errors in quantitative analysis:-

- Use of incorrect denominator when calculating the percentages
- Averaging percentages
- using a single average that distorts misrepresents the range of information

b) Qualitative analysis

Qualitative data consists of words and observations rather than numbers.

Analysis and interpretation of narrative data is often referred to as **content analysis**.

Sources of narrative data for analysis:

- Open ended questions and written comments on questionnaires
- Testimonials may give reactions to a program in a few words or lengthy comments either in person or in written correspondence.
- Individual interviews can produce data in the form of notes, summary of the individual's interview or word-for-word transcripts
- Discussion groups or focus group interviews often involve full transcripts and notes from a moderator or observer.
- Logs , journals and diaries might provide structured entries or free flowing text that you or others produce
- Observations might be recorded in your filled notes or descriptive accounts as a result of watching and listening
- Documents , reports and news articles or any published written material may serve as evaluation data
- Stories may provide data from personal accounts of experiences and results of programs in people's own words
- Case studies typically include several of the above.

Steps

1) Get to know your data

This means you read and re-read the text. If you have tape records, you listen to them several times and transcribe data. Write down any impressions you have as you go through the data.

Check the quality of the data. Is it complete and understandable? Is it likely to add meaning and value? Was it collected in an unbiased way?

2) Focus the analysis

Review the purpose of the evaluation and what you wanted to find out.

Based on your 'getting to know your data, think about a few questions that you want your analysis to answer and write them down.

You might focus your analysis by:

i) **Question, topic, time, period or event.**

In this approach you look at how all individuals or groups responded to each question or topic or for a given period or event. You organize data by question to look across all respondents and their answers in order to identify consistencies and differences

ii) **Case individual or group**

You may want an overall picture of:

- One case such as one family, or one agency
- One individual such as a teen participant in the program
- One group such as all teens 13-19 yrs.

Rather than grouping these respondents' answers by question or topic, you organize data from or about the case, individual or group and analyze it as whole.

You may want to combine the two approaches and analyze the data by both question and by case, individual or group.

3) Categorize information

This is sometimes referred to coding or indexing the data.

However, categorizing doesn't involve assigning numerical codes as you do in quantitative analysis.

To bring meaning to the words:

- i)* Identify themes or patterns- ideas concepts, behaviors, interactions phrases used etc.
- ii)* Organize them into coherent categories that summarize and bring meaning to the text. This involves reading and re-reading the text and identifying coherent categories.

You may want to assign abbreviated codes of a few letters, words or symbols and place them next to the themes and ideas you find.

Example:

3-13-02 Evaluation Workshop, Madison

Q 5. What do you need next to continue your learning about evaluation?

Line 7 is left uncoded because "Yes" is not usable data.

- | | | |
|---------|----|---|
| Trg | 1 | More advanced data analysis |
| T, R | 2 | More time/information on all the same concept |
| P | 3 | Just start doing them |
| Trg | 4 | Another workshop |
| Fdbk | 5 | Assessment feedback on how beneficial (or how well I did) the evaluation of my project was for USDA |
| Trg | 6 | How to measure long range planning outcomes |
| | 7 | Yes |
| P | 8 | Do it! |
| Trg | 9 | Need additional training and review |
| P | 10 | Practice makes perfect |
| T | 11 | Time to do actual reports |
| T, Fdbk | 12 | Time to complete an actual project report using this framework. Constructive feedback on strengthening that report. |
| Trg | 13 | More on how to decide evaluation reporting at the beginning of the grant or as you write the grant |
| P, M | 14 | Practice what I learned with assistance of a mentor |
| U | 15 | Unsure until I use the new ideas I've learned here. |
| P | 16 | Apply what I have learned |
| Trg | 17 | I need more training on analyzing data (need very basic hands on exercises) – maybe more exercises on indicators |
| Trg | 18 | Would have liked to go to observation interview training too! |
| R | 19 | More specific examples, i.e. completed logic models for different topics |
| U | 20 | Not sure yet. I'll know when I sit down to writing report. |
| R, P | 21 | Research suggested resources — develop more surveys – trial and error till I become perfect |

Provide a descriptive label (name) for each category you create and be clear on what you include in the category and what to exclude.

As you categorize data you might identify other themes that serve as sub categories. Continue to categorize until you have identified and labeled all relevant themes.

Ways of categorizing narratedata:

i) Using preset categories

You start with a list of themes or categories in advance and then search the data for these topics

These themes provide direction for what you look for in the data.

You identify the themes before you categorize the data and search for data for text that matches the themes.

ii) Emergent categories

You read through the text and find the themes or issues that recur in the data. These become your categories. They may be ideas or concepts you had not thought about. Sometimes you may combine these two approaches by starting with some preset categories and adding others as they become apparent.

Your initial list of categories May change as you work with the data.

Sometimes some sections of data fit into two categories. Therefore you need to create a way to cross index.

4) Identify patterns

Once you have identified the categories, you might:

- Sort and assemble all data by theme. It is important to capture the similarities and differences in peoples' responses within a category. It is helpful to write a summary for each category that describes these points.
- Sort and assemble data into larger categories- you may want to create larger super categories that combine several categories. You can work from more specific categories to larger ideas and concepts then see how the parts relate to the whole
- Count the number of times certain themes arise to show relative importance (not suitable for statistical analysis)

Show relationships among categories. You may also discover that two or more themes occur together consistently in the data. Whenever you find one, you find the other. You may decide that some of these connections suggest a cause and effect relationship or create a sequence through time

5) Interpretation

Use your themes and connections to explain your findings-attaching meaning and significance to the analysis.

Start by developing a list of keypoint or important findings you discovered as a result of categorizing and sorting your data.

Typical errors in qualitative data analysis

- Listing all narrative comments without doing any analysis
- Including information that makes it possible to identify the respondents
- Generalizing from comments to the whole group. Qualitative information seeks to

- Using quotes to provide positive spin. Consider your purpose of including the quotes. If you use quote:
 - Specify why you chose the selected quotes
 - Include enough of the text to allow the reader to decide what the respondent is trying to convey
 - Confidentiality and anonymity should be ensured
 - Get peoples' permission to use their words
 - Check with others about the value and usefulness of the quotes you select to include.

Statistics used in data Analysis

Statistics are used by the researcher to:

- Summarize and organized data
- Evaluate and interpret data
- Communicate numerical information

Statistics used include:

***a)* Descriptive**

These are statistics that organize and summarize numerical data gathered from samples.

They also help the researcher to examine the characteristics, behaviors and experiences of study participants.

These are categorized into:

***1)* Measures to condensed data**

These are statistics that are used to summarize and condense data.

***i)* Measures of central tendency**

These are statistics that describe the average, typical or most common value for a group of data.

Central - middle or average value

Tendency – this is the general trend of the numbers to cluster in a certain way. Measures of central

tendency include:

1) Mean

This refers to average sum of a set of values. It is appropriate for interval and ratio data. It is considered the most stable measure of central tendency if the distribution is normal. However it is influenced by extreme values.

2) Mode

This refers to the category or value that occurs most often in a set of data under consideration. That is the most representative category or value in the group.

It is the only measure of central tendency appropriate for nominal data. Category with the greatest frequency is called the **modal class**.

It is possible to have no mode.

Data can also have one or more modes.

- **Unimodal**- a set of data with one value most frequent.
- **Bimodal**- a set of data with two values/categories with the same high frequency
- **Multimodal**- a set of data with more than two values/categories with same high frequency.

3) Median

This refers to the middle score or value (50th percentile) in a group of data when data is arranged in ascending order. With interval or ratio data, median divides the frequency distribution of data in half. It is appropriate for ordinal, ratio, as well as interval data. Median is not influenced by extreme variables.

ii) Measures of variability

These describe how values differ among themselves in a distribution of values i.e. the distribution of scores around a particular central value or score and in this case the mean. They include:

1) Range

— This is the difference between the highest and the lowest value in a group of values or scores

It is a measure that can be used to gain a quick picture of the dispersion of the data. It is influenced by extreme values.

To correct this influence, inter-quartile range may be reported.

2) Percentile

This is a datum point below which lie the values in a frequency distribution. E.g. 80th percentile of a score means 80% of the subjects received lower score or 20% of the subjects higher score.

3) Standard deviation

This describes how the values range from the mean of the distribution. It involves subtracting the mean from each score to obtain the deviation. It is most commonly used when interval or ratio data is obtained.

4) Variance

This Standard deviation squared.

iii) Measures of relationship

These are concerned with the correlations between variables i.e. how values of one variable are related to the values of another variable.

Relationship is examined through:

1) Correlational coefficients

This determines the magnitude and direction of the relationship between two variables. Generally, correlational coefficients are calculated on measurements obtained from each subject on two variables

2) Scatter plots/scatter diagrams/scattergram

This is a Graphic representation of the relationship between two variables. The X variable is plotted on the X-axis and the Y variable on the Y-axis. Dots are used to indicate where X and Y intersect. The concentration of the dots in the graph indicates positive/negative correlation.

3) Contingency tables

If data are nominal or categorical, relationship cannot be depicted on a scatter plot since no actual results are available. It is only the frequencies of occurrence of the values are presented. Contingency table/cross tabulation table is a means of displaying the relationships between sets of nominal data. E.g. relationship between gender and level of motivation

b) Inferential statistics

These are statistics that are used to make decisions or inferences about a population. They are used to:

- i) Estimate the population parameters from the sample statistics. This involves determining the confidence intervals.

ii) To test hypotheses

All inferential statistical tests are based on the assumption that chance is only explanation for the relationships that are found in research studies. A researcher wants to demonstrate that chance is not the reason for the relationships found in research.

Choosing a statistical test

There are two types of inferential tests

- Those that search for differences in sets of data. These are used when you want to determine if there is a significant difference between groups
- Those that search for correlations between sets of data. These are used when you want to determine if there is a significant correlation between variables within one group.

Choice of the statistical test is based on:

- The hypotheses for the study or the research questions.
- Sample size
- Number of groups or sets of scores to be compared
- Whether the scores / observations are dependent or independent
- Whether you are comparing groups or sets of scores or correlating variables
- The number of observations available for each group.

Inferential statistics can be classified as:-

i) **Parametric**

These are concerned with population parameters and thus make assumptions about the population from which a sample was drawn.

For a parametric test to be used it is assumed that:

- o The level of measurement of data is interval or ratio
- o Data is taken from population that are normally distributed
- o Data are taken from population that has equal variances on the variable being measured

Examples: t-test

ii) Non parametric

These do not make assumptions about the population from which a sample was selected.

These are used

- With nominal or ordinal data.
- When sample sizes are small Examples: wilcoxon's

DATA PRESENTATION

a) Narrative presentation of findings

The findings of a study should be clearly and concisely presented in the text

Certain information should always be included in the text when discussing the study hypothesis:

- Statistical test that was used
- Test results
- Degrees of freedom (df)
- Probability value (p-value)

In qualitative research studies, hypotheses are rarely tested. Therefore inferential statistics are not included in these reports. Narrative presentation frequently contains a lot of direct quotes made by the subjects. Then the researcher may present a summary of the findings by discussing patterns and themes found in the data.

b) Tables

These are means of organizing data so they may be more easily understood and interpreted.

Guidelines concerning tables

- Tables should appear as soon as soon as possible in the report after they have been referred in the text
- Information presented in tables should also be discussed in the narrative of the report
- Titles should be clear, concise and contain the variables that are presented in the data. The name of the statistical test should not be used
- All data entries should be rounded to the same number of decimal places (1-2 commonly used)

- Decimal points if present should be lined under each other in the data columns
If data are not available for a section of the table, a dash (-) should be entered rather than leaving a blank space, to make it clear that no data have accidentally been left off the table

Examples of tables

1) Frequency distribution tables

Frequency distribution is appropriate for reporting all levels of data. All values or scores are listed and the number of times each one appears is recorded. Values may be listed from the highest to the lowest.

This is depicted using a frequency distribution table. Frequency distribution table can be:

i) Simple table

A single line of characters explain in a single column of information. It shows the distribution of scores in a sample for a specified variable. Size of the frequencies should always be equal to the sample size "n"

ii) Compound table

A single line of characters has been described by two or more compounds of information

Grouped frequencies

When the range of scores is large, it may be helpful to group the scores into smaller before counting the frequencies. Other instances when data is grouped include when:

- The scores are distributed in such a way that certain scores are not obtained by any subject
- The sample is very big since the frequency would be too long
- Information sought is sensitive such as annual income, age and the response categories in the questionnaire are given in intervals

Class intervals normally start with the lowest score as a multiple of the interval width.

Interval width refers to the size of the class interval. Class interval depends on the sample size and range of scores but should be between 10-15 in number.

Principles governing conversion of raw scores into group score:

- All intervals should be the same width
- Intervals should be continuous throughout out the distribution i.e. even if there are no scores in a particular class interval, the class interval must be retained and a frequency of zero indicated against it.

- Too few intervals lead to loss of accuracy and too many class intervals result in inconveniences and range between 10-15
- Intervals must be exhaustive and mutually exclusive.

c) Figures

Term used to indicate any type of visual presentation other than a table.

Figures may help enliven a narrative presentation since they have a visual appeal and should be considered as a valuable means of displaying research results

Figures include:

1) Graphs

Graphical representation of a frequency distribution is a valuable supplement to statistical analysis. The graph enables the reader to see the trend of distribution more easily than is possible by looking at numbers in a frequency distribution.

Types of graphs commonly used to present data include:

i) Bar graph

Figure used to represent frequency distribution of nominal data and some type of ordinal data

It is especially useful when the categories of variables are qualitative rather than numerical. The length of the bar represents frequency of occurrence of the category. To show that data being presented are separate categories, the bars do not touch each other. Data are presented on only one variable in a bar graph

ii) Histogram

Graph that use bars to represent frequency distribution of a variable measured at the ordinal, interval and ratio.

Data are presented on one variable in a histogram

Bars are of equal width and touch each other to indicate that data are being presented in a continuum

Width of the bar represents the size of the class interval

The height of the bar represents the frequency of occurrence of each class interval

2) Line drawings

i) Frequency polygon

This is a graph that is obtained by joining the midpoints of class interval. The midpoint is established by summing up the lower and upper class limits of each class interval and dividing by 2. A dot is placed above the midpoint of each class interval. These dots are connected.

It is used for grouped data. It represents the frequency distribution of ordinal, interval or ratio data.

3) Charts

Figure used to present data but data has to be converted. The mostly commonly used is the pie-chart.

4) Photographs

In form of pictorials.

d) Percentages

This is a statistic that represents the proportion of a subgroup to a total group expressed as a percentage ranging from 0-100

Minimum number for computation should be at least 20

COMMON ERRORS THAT ARISE DURING DATA ANALYSIS.

Type I error: - it is when a null hypothesis is rejected when it is actually true; a decision is made that a relationship exists between variables when it does not.

Type II error: - a decision is made not to reject the null hypothesis when it is false; a decision is made that no relationship exists between variables when, in fact, a relationship does exist.

BIAS DURING DATA ANALYSIS.

- **Response bias:** - it is the tendency of respondents to distort their responses.
- **Social desirability response bias:** - refers to the tendency of some individuals to misrepresent their responses consistently by giving answers that are congruent with prevailing social values.
- **Extreme response rate:-** results when a person characteristically endorses extreme responses alternatives e.g. strongly agree / disagree
- **Acquiescence :-** which is yea-sayer's tendency to agree with statements regardless of their content
- **Nay –sayer's:-** disagree with most statement

Common errors of interpretation which need to be avoided:

- Failure to see the problem in proper perspective
- Failure to appreciate the relevance of various elements
- Failure to recognize limitations in the research evidence
- Misinterpretation due to unstudied factors
- Ignoring selective factors
- inadequate attention to individual cases

ASSUMPTIONS AND LIMITATIONS

ASSUMPTION

This refers to any fact that a researcher takes to be true without actually verifying it. Example:

- The study assumes that the information that will be provided by the respondents will be accurate and reliable
- The health service providers have kept proper records that will provide the information required in this study.

Assumptions provide the reader with vital information which influences the way the results of the study are interpreted.

LIMITATIONS

This refers to any aspect of the study that may influence the results of the study negatively but over which the researcher has no control over.

Common limitations include:

- Scope of the study which may not allow generalizations
- Sample size
- Respondents withholding some information especially with sensitive issues.

The researcher should not include factors that can be controlled unless a good rationale is provided.

98 / Page

APPENDICES

- Appendices are materials that may be of interest to some but not all readers of

which are not sufficiently pertinent to be included in the body of a proposal.

- They are materials which are not essential to understanding the study.
- Usually comprise of instruments, maps, letters of introduction & data, personnel met, research approval letters, budget, time frame work & explanatory notes.
- They are labeled appendix 1, 2, 3 or a, b, c.

a) BUDGET

Is the financial implication of carrying out the research. Should be clear & reasonable.

It should consider the following:-

- ❖ Personnel or field assistants
- ❖ Travel & accommodation
- ❖ Stationery
- ❖ Equipments/materials
- ❖ Consultancy e.g. for data analysis if required
- ❖ Subsistence
- ❖ Services e.g. secretarial, photocopying, printing, binding
- ❖ Miscellaneous- usually 1% of the total cost.

The budget is usually presented in tabular form i.e.

Table 7: Sample of a Research Budget

NO.	ITEM	DESCRIPTION	AMOUNT
1.	STATIONERY	• 10 REAMS OF PHOTOCOPY PAPER @ 400/= • 2 PACKETS OF DISKETTE @ 500/=	4000/= 1000/=
2.	PERSONNEL	• 5 FIELD ASSISTANTS @ 500/= PER DAY FOR 14 DAYS	25,000/=

b) TIME FRAME WORK (WORK PLAN)

99 / Page

This is the schedule/ time-table of the activities over a period of which the research is to be carried out. It's presented in a tabular form e.g.

PHASE/ACTIVITY	TIME/MONTHS	DATES
Development of proposal	3	June-Aug 2010
Developing & piloting instruments	1	Sept.2010
Data collection	1	Oct.2010
Data organization, analysis & interpretation	2	Nov.- Dec.2010
Typing/editing/report writing/submission	3	Jan. – mar. 2011

N/B: - If researcher suspects some external factors may prevent him/her from following the time plan he should state those factors immediately below the table as obstacles to time tabling.

DISCUSSION OF FINDINGS, CONCLUSION, IMPLICATIONS AND RECOMMENDATIONS

DISCUSSION OF FINDINGS

This is the more subjective section of the research report.

It allows the researcher to make interpretations of findings. Findings are interpreted in the light of the theoretical framework and literature review.

In discussing the findings, the researcher discusses the aspects of the results that are in agreement and that are not in agreement with previous research and theoretical explanation. However no new literature should be introduced that is not already cited.

The inferential statistics used to test hypotheses should also be presented. Whether the null hypotheses is rejected, and research hypothesis supported or rejected and the results are in the opposite direction from the prediction of research hypotheses should also be discussed.

CONCLUSION

The study conclusions are the researcher's attempts to show what knowledge has been gained by the study and also an attempt to generalize the findings.

Conclusion is written in view of the study problem, purpose, hypotheses and theoretical framework.

In drawing conclusions the researcher should take into consideration

- The sample size,
- Population from which the sample was drawn
- The study design.

Conclusions are more abstract and use general terms as opposed to findings of the study that are concrete and tied to specific data. Conclusions go beyond the findings

Conclusions are also tentative hence the use of the word “appears” since we cannot prove.

IMPLICATIONS

Implication section deals with what changes are suggested based on the conclusions. The implications should contain the “should” that result from the study.

A study implication might be

- No change is needed
- More research is called for to further verify the study results
- That changes need to be made based on study conclusions

Implications may be addressed to the clinicians, educators, researchers, administrators, theorists.

RECOMMENDATIONS

This is the last section of the research report. This section should contain recommendations for further research. This section should propose:

I. Replication of the research study

- 1) This involves carrying out a study similar to one that has previously been done. Only minor changes are made from the previous study such as using a different type of sample or setting.

Partial replication is more common where an instrument is changed or a new tool is added.

- 2) A new study in which the present study limitations are considered. This may involve:

- Alteration in the sample e.g. different age group or educational level
- Alteration in the instrument e.g. change an existing tool or use another instrument)
- Control of variables e.g. taking only a certain age group in the sample or selecting a random sample instead of using a convenience sample)

- 3) Change in methodology (using structured observation instead of self-report

methods)

II. Extension of the research study

WRITING A RESEARCH REPORT

A research report is a question answering or problem solving document. A research report informs the reader of the problem the researcher initially set out to investigate, the method of investigation used and the researcher's findings.

The report should present data fully and adequately and it should include accurate interpretations of the analyses of such data and should relate findings back to the objectives, hypotheses or research questions. It should therefore be formal, precise and economical, consistent, and orderly etc.

A well written report comprises:

1. Title

A title should be simple yet informative.

Functions of title

- It gives the reader some initial information about the area of research report has covered and therefore a pointer to the content of the report.
- It is used for indexing once the document becomes official. People interested in the document may use the title to search for the report in a library or document centre.

A title page should have:

- Title of the project
- Investigators/financiers/supervisors
- Addresses.

2 Dedication

Some authors of research reports dedicate their work to a person or persons whom they deem special in their lives.

3 Acknowledgement

It is a matter of courtesy for the researcher to acknowledge all the individuals, institution and /or organizations that helped in the course of research. Help may include: time, resources, information. Mention only a few individuals by name to keep the acknowledgement section as short as possible

4. Abstract

This is a research report that is placed before the full report and briefly summarizes or highlights the major points. It is a reflection of the salient findings of the research study. It should be placed in front of the report preferably after the dedication but on a separate page.

A good abstract should present the reader with:

- Introduction of the topic
- Purpose/ rationale of the study
- Study objectives
- Study methodology
- Major results and conclusions of the study.

An abstract may or may not be structured. It is usually written at the end and should be approximately 300 words

5. Table of contents

This is like a map that guides the readers in locating various sections of the research report.

It contains the chapter headings, main headings, subheadings and corresponding page number of each in the body of the document.

It helps the reader to locate the desired sections quickly and easily.

6. List of tables

A researcher uses tables to summarize information in a logical format or sequence. The number and title of each table appearing in the body of the report is listed together with the corresponding page number

7. List of figures

A figure is any pictorial representation used to clarify specific points in a discussion. They include graphs, charts, diagrams and photographs.

Give the number and title of the figure and the page number.

8. List of abbreviations and acronyms

Abbreviation is a short form of word e.g. Dr. Prof.

Acronym is a contraction formed by taking the first letter of several words e.g. WHO, MOH

- Write in full the first time that the acronym is used.
- Do not abbreviate military, religious or political titles.
- Abbreviate units of measurements only when they are used often in a report.
- Use only abbreviations that your audience will understand
- Do not abbreviate days of the week or months
- Abbreviate time designations only when they are used with actual time
e.g. 4.30 p.m.

9. Operational definitions.

This refers to definitions /meanings of words that are used in the study. Use WHO definitions

10. Introduction (CHAPTER 1)

The purpose is to provide a basis for the study, it provides background & the setting required to put research problem in proper context.

This has several components:-

- Background of the study - It introduces the status and the theoretical framework of the study. It's the rationale of the study providing evidence and conditions of the existing situation to make the reader feel the urgency of the problem and to study it, to contribute to its solution. It portrays the **history & character of the problem**.

It has 3 main dimensions- conceptual, theoretical and contextual dimensions.

- ❖ Theoretical describes theories on which the study will be based. It basically describes the relationship between the variables in ideal situations.
- ❖ Conceptual provides definitions of terms in the topic.
- ❖ Contextual describes problems in the context of the study.
 - The problem statement
 - Study objectives
 - Hypotheses and /or research questions
 - Assumptions of the study
 - Limitations of the study

The major role of the introduction section is to outline the gap or gaps that exist in the area of study and present the rationale of the study.

This leads to the statement of the problem. The researcher should quote existing studies and theories to support the problem statement. Literature cited in this part of the document should be limited to the most relevant and current.

11. Literature review (CHAPTER 2)

This is a researcher's critique of findings from other studies done in a related area.

The critique should include an assessment of methodologies used in these studies, theoretical or conceptual frameworks and the relationships or differences between the researcher's study and the studies reviewed.

12. Methodology (CHAPTER 3)

This gives details regarding the procedures in conducting the study.

It includes:-

- Research design
- Research site
- Study population including inclusion and exclusion criteria
- Sample size determination
- Sampling techniques
- A description of methods and instruments/tools used to collect data. Pretesting of the tool should also be discussed.
- Data collection process including training of Assistants.
- Quality assurance procedures-calibration, timing etc.
- The measurements of variables and techniques to be used in data analysis including statistical procedures that will be used.
- Limitations and assumptions
- Ethical considerations,

13. Results and discussions (CHAPTER 4 & 5)

The purpose of this part is to present the results of the data in a systematic way. If the study is empirical in nature, the researcher uses statistics to summarize the results and make generalizations on the population

Basic principle in presenting results is to give all the evidence relevant to the research objectives and questions if any.

When writing the results and discussions, it is advisable to start with a short introduction that describes the general procedures followed in analyzing the data

Pointers in writing this section include:

- Present the descriptive results of your analysis first. Descriptive statistics include: frequencies, percentages, means, modes, correlations. These statistics should be presented in tables to make the document user friendly.
- It may not be possible to report every result of analysis. The researcher therefore selects the most important findings for reporting. This should be guided by the study's objectives or hypotheses as stated in introduction part
- Not every result or table generated in the course of the analysis or table generated in the course of analysis is presented in the report. Objectives and hypothesis should guide the researcher.
- All tables used should be numbered systematically and given appropriate titles
- Discussion based on the information contained in a table should be placed either immediately before or after the table
- Graphs, charts and other visual presentations should be included in the results and discussion section if they add value to the content. Sometimes tables and other graphic representation of the data are placed in the appendices

14. Summary, conclusions and recommendations (CHAPTER 6)

Summarize the study undertaken, conclude and make recommendations

15. References

Use the APA referencing system.

16. Appendices

This contains information that the researcher doesn't deem necessary to include in the body of

the report. This includes:

- Instruments used in the study
- Consent forms
- Correspondences related to the study such as letters of approval to conduct the research
- Copy of the research permit and terms of reference if any
- Statistical tables used for data analysis.

DISSEMINATION OF STUDY FINDINGS

Once the research has been conducted, the findings should be disseminated as widely as possible.

Ways of dissemination

i) Presentation of findings to professionals

This involves paper presentations in seminars, conferences, discussion forums, workshops and meetings.

ii) Presentation to the world community through:

- Media
- Workshops
- Community discussions

iii) Presentation to specific stakeholders like Government departments, NGOs and Research community.

ROLES OF A NURSE IN DISSEMINATING RESEARCH FINDINGS

1. Selecting a communication outlet: - decide on the method of communication.
2. Knowing the audience: - good research communication depends on providing information that can be understood by consumers.

3. Developing a plan: - before beginning to prepare research reports researchers should develop a plan, part of the plan involves decision about the communication outlet and the audience for the report.
4. Writing effectively.

TYPES OF RESEARCH REPORTS.

There are four major kinds of research reports

- I. Thesis and dissertations
- II. Traditional journals articles
- III. On line report
- IV. Presentations at professional meetings where we have two methods oral presentation and poster presentation

RESEARCH CRITIQUE

This is done by experts before it's published.

It's done by looking at the following:-

- Researcher qualifications, title, .abstract, introduction, purpose, problem statement, literature review, theoretical/conceptual framework, assumptions limitations, hypothesis, definition of terms, research design, setting, population and sample, data collection method, data collection instrument, validity & reliability of the instrument used, data analysis, discussion of findings, conclusion, implications, recommendations, ethical consideration, other considerations, summary.

BARRIERS TO NURSING RESEARCH UTILIZATION.

- Nurses lack of knowledge.
- Nurses negative attitudes towards nursing research.
- Inadequate means of disseminating nursing research findings.
- Lack of institutional support.
- Study findings that are not ready for use in nursing practice.

HOW TO IMPROVE NURSING RESEARCH UTILAZATION.

The following steps should be followed

- A. Identify specific clinical nursing practice problems
- B. Assess the existing knowledge base on the issue
- C. Design a nursing practice innovation based on a scientific research base
- D. Implement clinical trials and evaluation of the innovation
- E. Decide to adopt, revise, or reject the innovation
- F. Develop strategies to extend the innovation to other settings
- G. Determine means of continuing the innovation over time.